

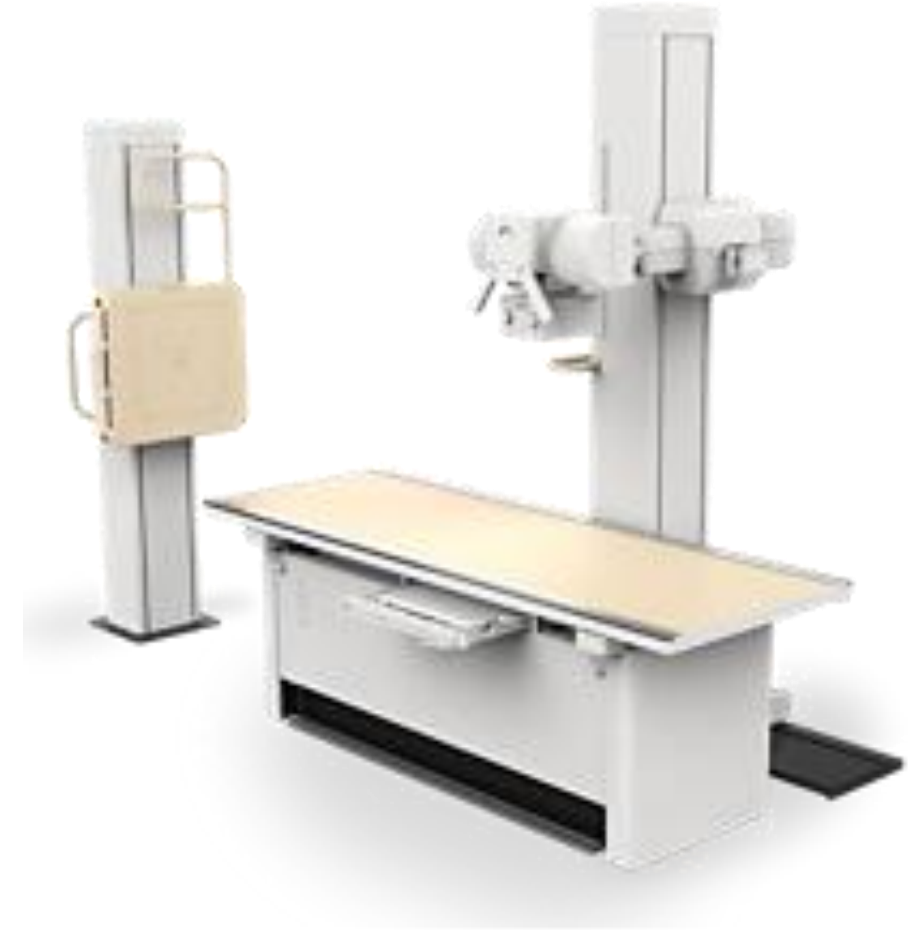
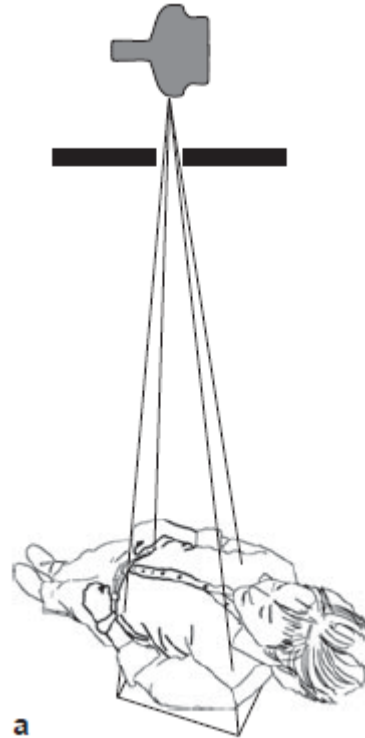
X-Ray based planar radiography
and computed tomography (CT)

X-ray Planar Radiography

- 2D
- Analog and Digital recording available
- Portable

Key aspects:

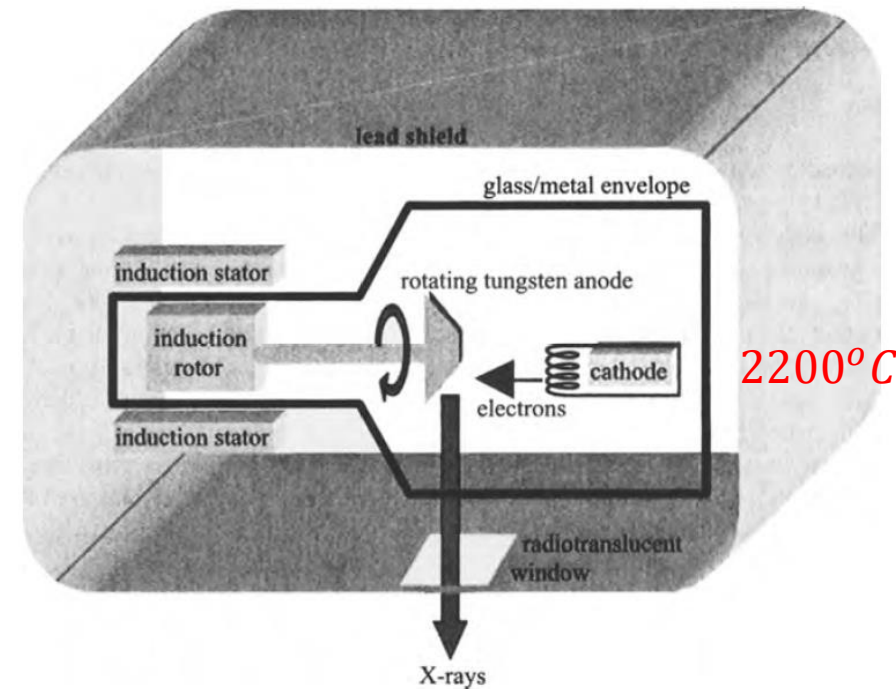
- X-ray generation (the X-ray Tube)
- X-ray interactions with the body
- X-ray detectors
- Factors affecting Resolution
- Signal to noise ratio
- Contrast



X-ray Tube

Peak potential difference between the Cathode and the Anode– is usually between 25 kV to 140 kV

- Thermionic emission of electrons from the Cathode forms a tube current that hits the anode which is at a positive potential (KVp)
- Both Cathode and Anode are usually made of Tungsten
 - High thermal conductivity
 - High melting point
 - Low Vapor Pressure
- Bevel angle is to achieve a small focal spot size



F is width of electron beam

$$f = F \sin \theta,$$

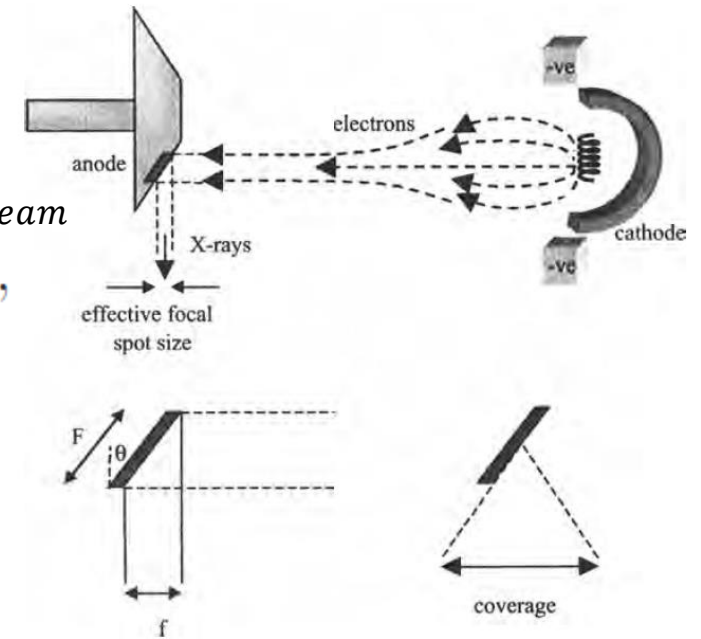


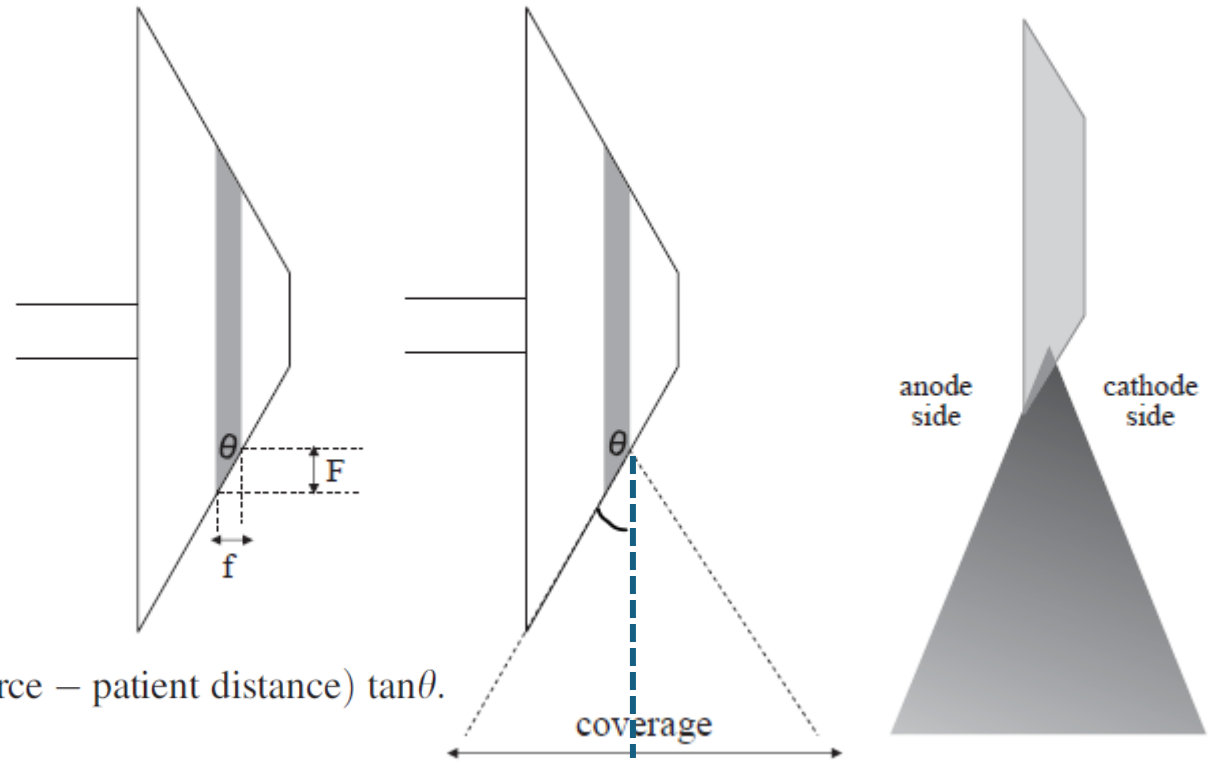
Figure 1-5: The Rotating Anode



The focal track (i.e. the location of the electron interactions) can be seen as the darker gray stripe ringing the edge of the anode disk.

Molybdenum is also used for Anode

$$\text{Coverage} = 2(\text{source} - \text{patient distance}) \tan \theta.$$

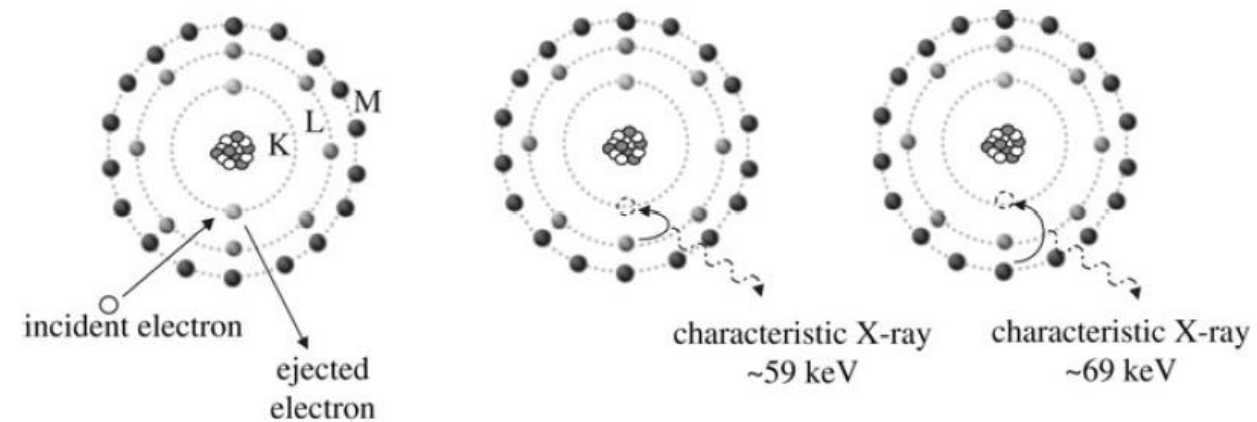
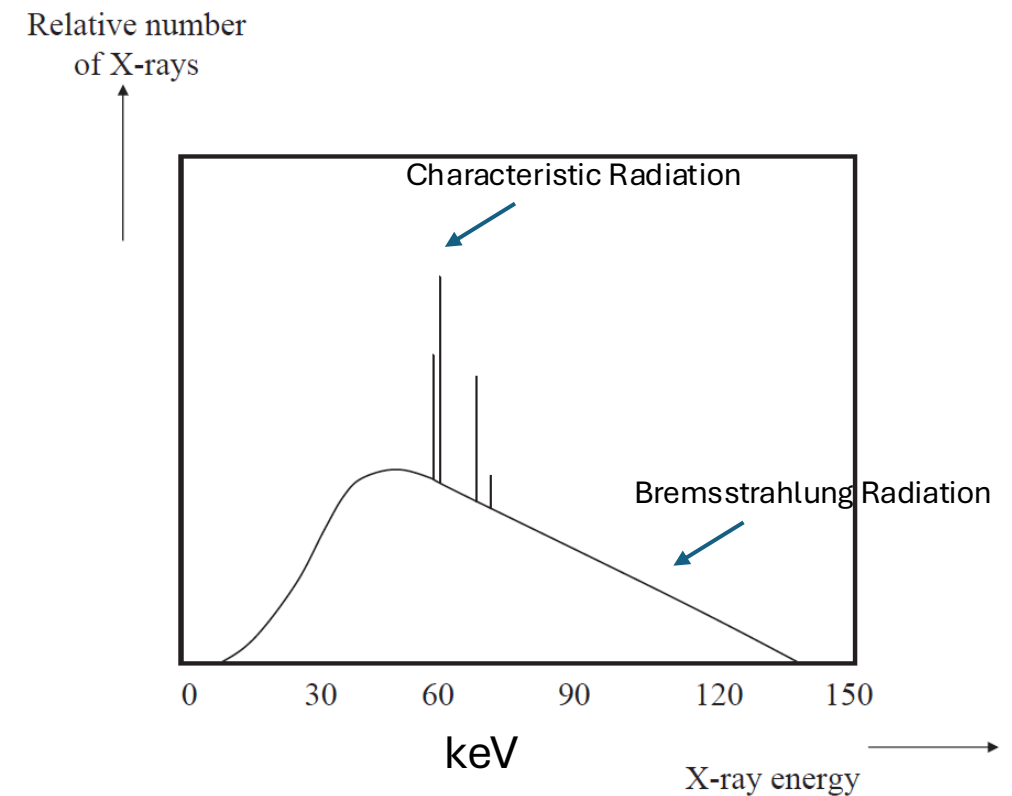
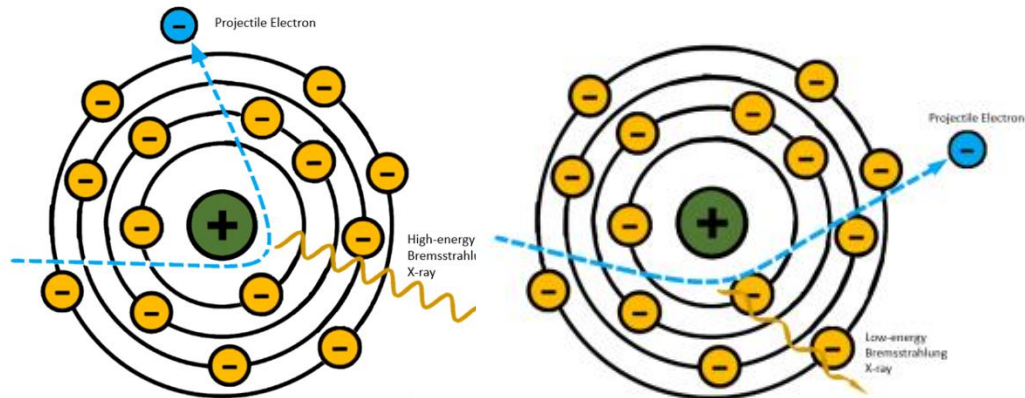


Heel effect: X-ray intensity on the cathode side is higher than that on anode side as the rays pass longer distance within the anode material on the anode side.

- **User controlled parameters** - kVp, tube current, and exposure time, based on the application.
- Tube-current range – 50mA to 400 mA for X-rays and 1000mA for CT.
- Heat-dissipation in the Anode is the limiting factor for achieving high tube-currents.

Mechanisms leading to X-ray radiation

- Bremsstrahlung Radiation (Braking Radiation)
 - Deceleration of the electrons within the anode material due to the nuclear-field
- Characteristic radiation (specific frequencies, characteristic of the anode material):
 - Radiation resulting from electron transitions from a higher energy shell (L or M) to a lower one. The transition occurs because holes are created when an incident electron (from the tube current) knocks an electron off from a lower shell an atom in the anode.



X-Ray interactions within the tissue

- Photoelectric attenuation
 - Energy of the incident X-ray is lost in entirety in the tissue, as it gets used up for knocking off electrons usually from low energy shells of the atoms in the tissues (Carbon, Oxygen, Calcium etc.)
- The probability (P_{pe}) of a photoelectric interaction occurring for an x-ray within the tissue depends on its
 - Energy (E)
 - Effective Atomic number of the tissue (Z_{eff})
 - Density of the tissue (ρ)

$$P_{pe} \propto \rho \frac{Z_{eff}^3}{E^3}$$

In generally, about 90% or more of the incident X-rays are absorbed in the body

EFFECTIVE ATOMIC NUMBER AND ENERGY ABSORPTION IN TISSUES*

By F. W. SPIERS, Ph.D.
The General Infirmary, Leeds

1946

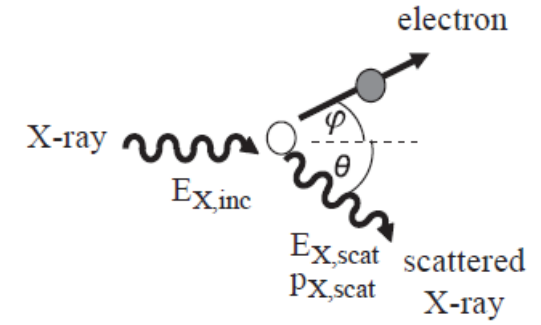
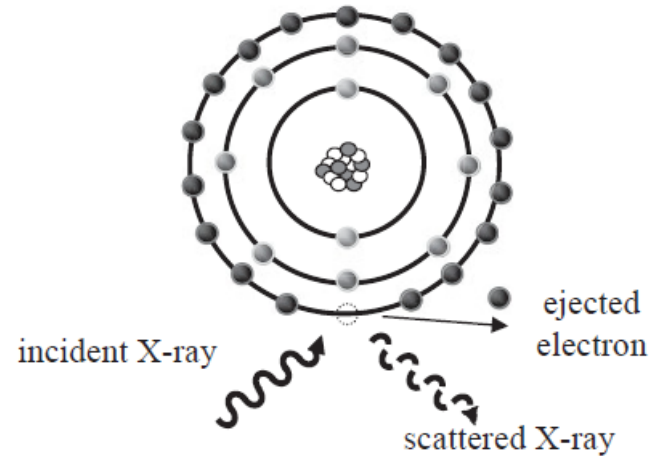
(1) Tissues	(6) $\bar{Z}$	(7) ρ gm./c.c. (
Muscle I (biceps)	7.42	1.00
Muscle II (biceps)	7.46	1.00
Muscle III* (gastrocnemius)	7.2	0.98
Skin	7.31	1.00
Subcutaneous fat	5.92	0.91
Animal fat* ..	5.6	0.90
Bone I (femur) · (as powder)	13.8	1.85
Bone II (femur) (cut section)	14.0	1.87
Bone III (animal) (as powder)	13.9	1.95

X-Ray interactions within the tissue

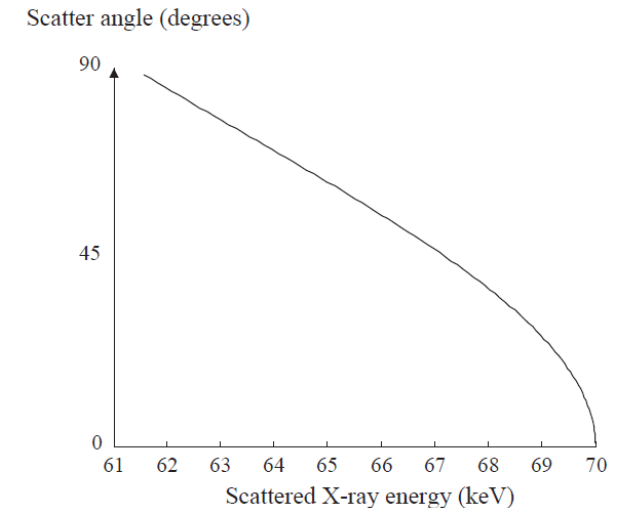
- Compton Scattering
 - Incident X-ray knocks off the electrons in outermost/valence shells and in the process loses some energy and deviates (gets scattered) from its path by an angle θ .
 - $\Delta\lambda$ between the incident ray and the scattered ray is given by

$$\Delta\lambda = \frac{h}{mc} (1 - \cos \theta)$$

$$\Delta E = \frac{E_{X,inc}}{1 + \frac{E_{X,inc}}{mc^2} \cdot (1 - \cos \theta)}$$



Electron density of the tissue is the other major factor that affects Compton Scattering



X-Ray interactions within the tissue

- Intensity of the X-rays is defined as the ‘power incident per unit area’.
- Power, in turn, depends on the total # of X-rays produced and the energy of the X-rays.
- The # of X-rays produced is proportional to the tube current.
- Energy of the X-rays produced is proportional to the square of the accelerating potential kVp.

$$I \propto (kVp)^2 \cdot (mA)$$

Radiation dose to the patient is proportional to the intensity of the X-rays and exposure time

X-Ray interactions within the tissue

- Attenuation of the X-rays as a function of distance, follows an exponential relationship (Beer Lambert Law)

$$N = N_0 e^{-\mu(E) \cdot x} \quad I = I_0 e^{-\mu(E) \cdot x}$$

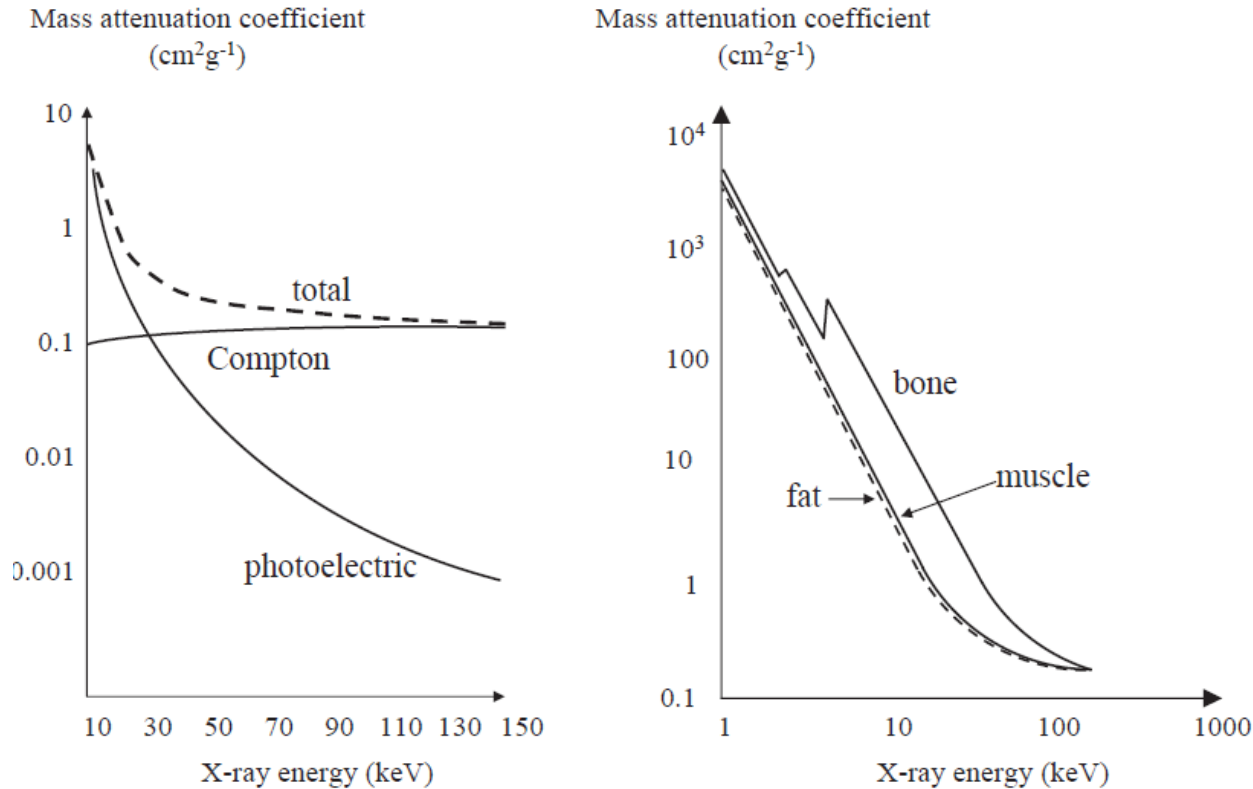
N_0 is the number of incident X-rays, $\mu(E)$ is the linear attenuation factor, which is dependent on the energy of the incident X-ray. μ is measured in (attenuation per cm^{-1} ($\mu = \frac{\ln(\frac{N_0}{N})}{x}$))

μ depends on the incident energy and the tissue type

$$\mu = \mu_{photoelectric} + \mu_{Compton}$$

μ/ρ – mass attenuation coefficient; units of cm^2/g

X-Ray interactions within the tissue



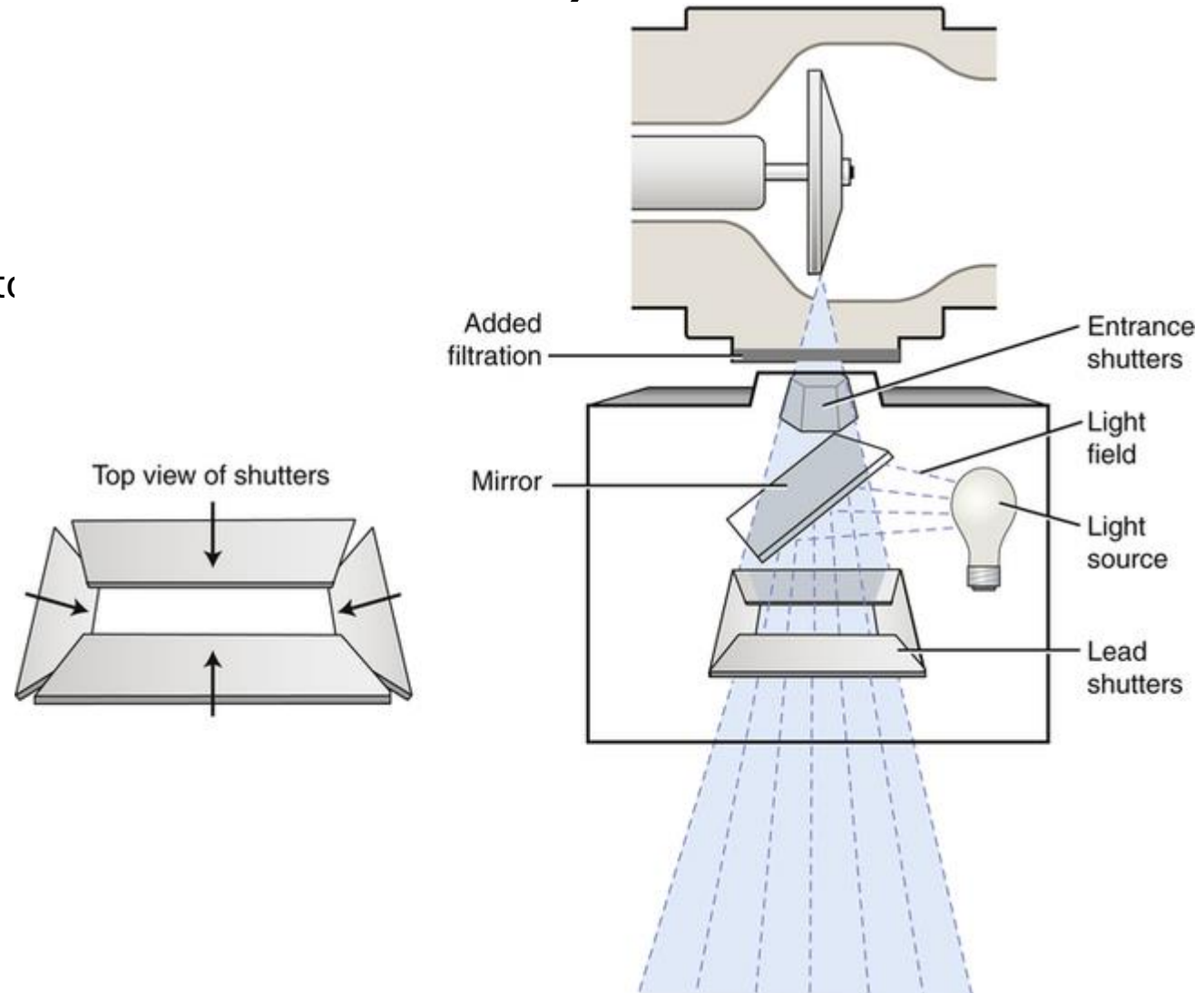
- At lower energies, absorption is high ($\frac{\mu}{g}$ is high)
- As energy of the incident X-ray increases, μ/g decreases
- The probability of photoelectric attenuation increases as the energy approaches the binding energy of the k-shell, there is an abrupt jump in the value of μ/g (i.e., increased absorption), called the *K-edge*.
- Another way of characterising X-ray absorption in a tissue is: Half Value Layer (HVL), - the tissue thickness at which the intensity of the incident X-ray reduces by $\frac{1}{2}$.

- $HVL = \ln(2) / \mu$

Example...@100 keV HVL of muscle is 3.9 cms

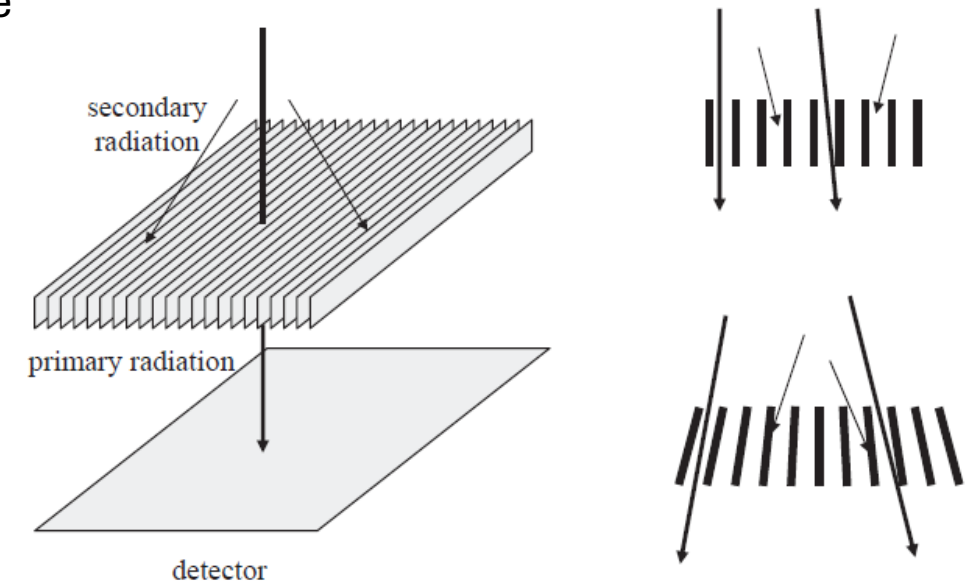
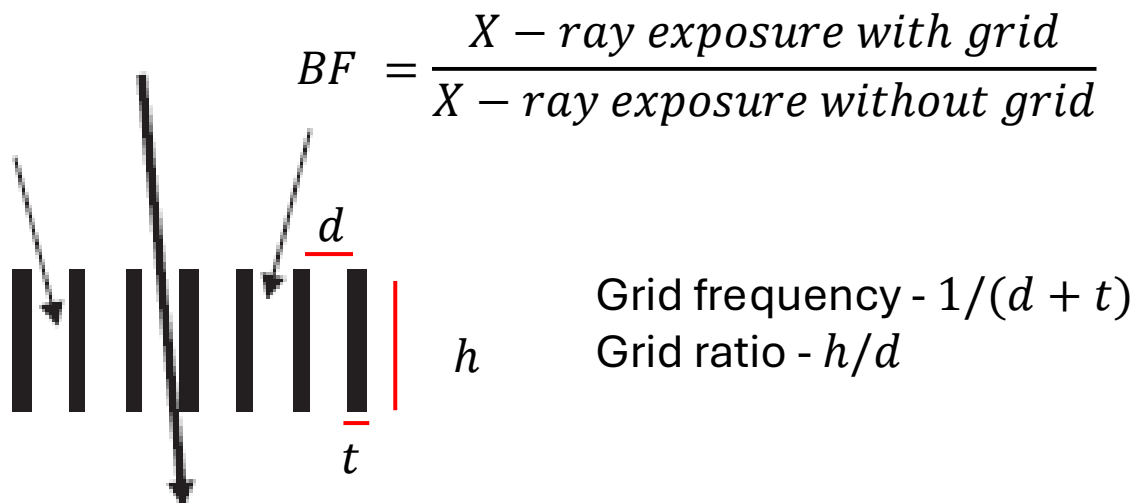
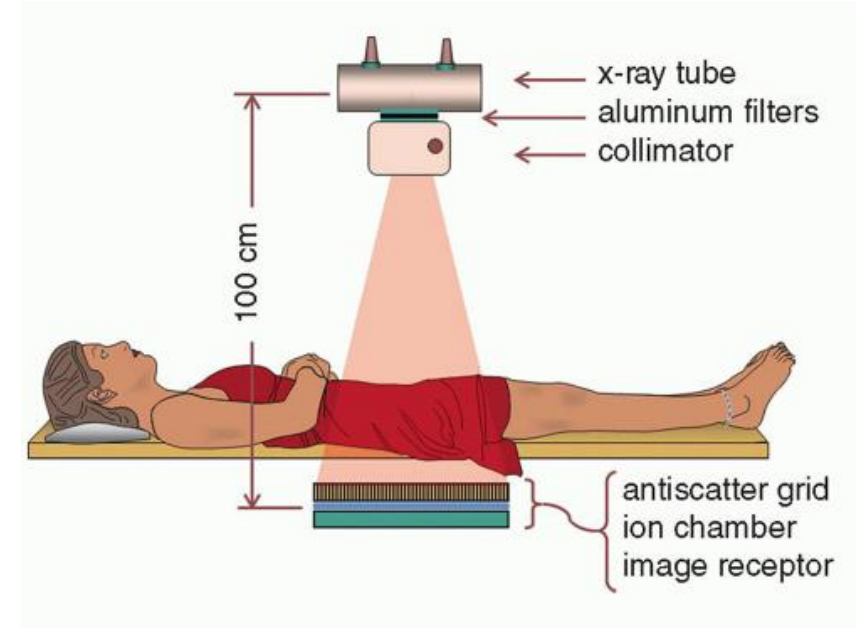
X-Ray Collimation (beam restrictor)

- Coverage is determined by the bevel angle.
- However, further restriction of the X-ray beam to the anatomy of interest (field of view), called collimation, is performed using lead-window
- Collimators or beam restrictors are placed between the beam source and the patient
- This reduces patient dose
- Also reduces scatter



X-Ray Anti-scatter grid

- Placed between the patient and the X-ray detector, it restricts the scatter radiation from reaching the X-ray detector/film.
- Reduces SNR (which is proportional to the # of x-rays reaching the detector)
- To compensate for this, Intensity is usually increased to obtain good SNR image.
- The factor by which patient dose gets increased, due to the use of an anti-scatter grid is called the **Bucky Factor**. (typical BF 4 to 10)





Without grid

*With grid and no increase in
patient exposure*

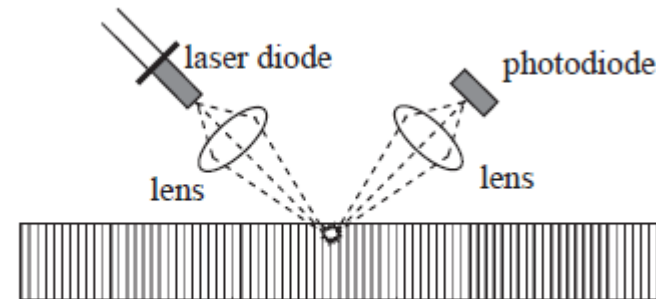
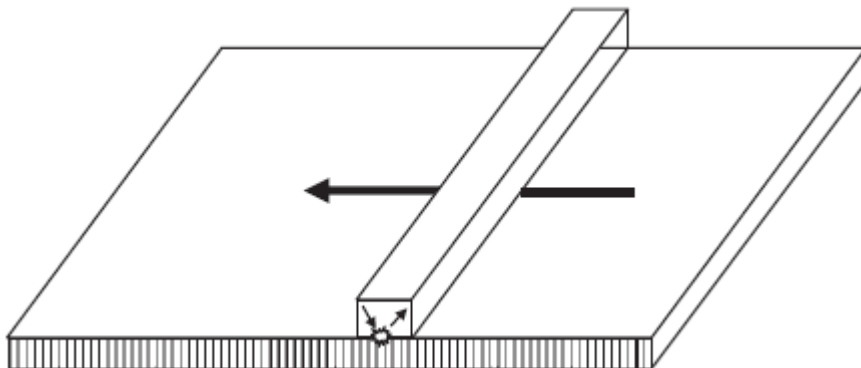


X-Ray Detectors

- Computed Radiography
- Digital Radiography

Computed Radiography detector

- Contains a **detector plate** (e.g. BaFX:Eu²⁺ where X is either Iodine or Bromide) which release electrons that are stay trapped within the material for minutes to hours.
- When the detector layer is exposed to optical laser, it causes most of these electrons to return to ground state, which in the process leads to emission of visible light. This is then detected using an array of **photo-diodes** that convert this to a voltage which is then amplified and digitized.
- Sampling can happen every 50 μm
- Dynamic range is large – spanning 4 orders of magnitude



X-Ray Detectors

- Computed Radiography
- Digital Radiography

Digital Radiography detectors

- Two types – Indirect and Direct
- Indirect: Uses scintillators (CsI:TI – Cesium Iodide doped with Thallium), which are materials that emit visible light when exposed to X-rays. They are used in combination with a 2D flat panel detectors, consisting of an array of photodiodes, a capacitor, and Thin Film Transistors.
- Direct: Uses detector materials (amorphous Selenium) that converts absorbed X-rays to electrical signal.

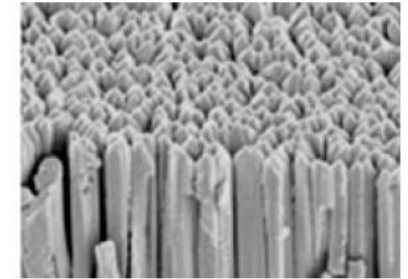
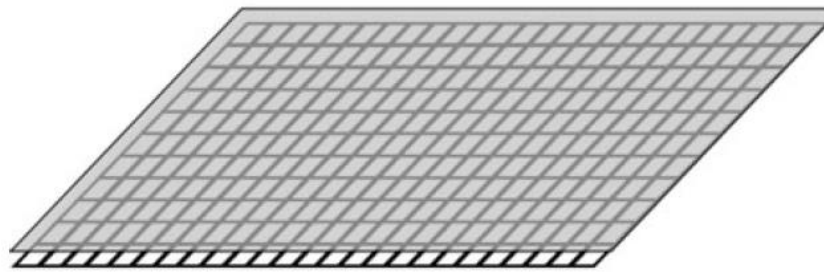
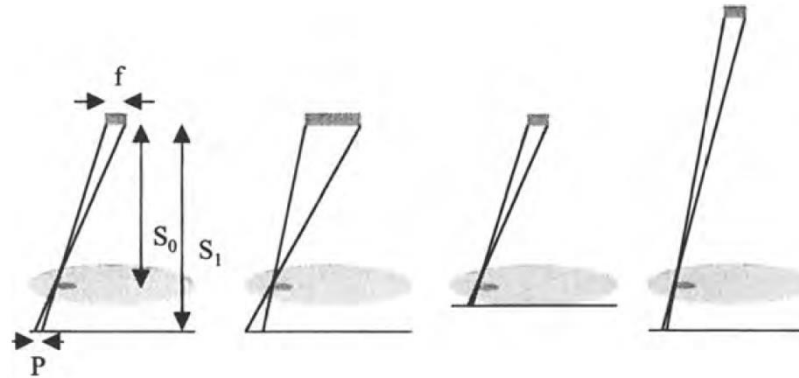


Figure 2.16

(left) A thin CsI:TI layer (shaded) placed on top of a SiH active matrix array. (right) An electron micrograph showing the needle-like structure of the crystals of CsI:TI.

Some additional factors affecting X-ray Image quality

Effect of focal spot size



object

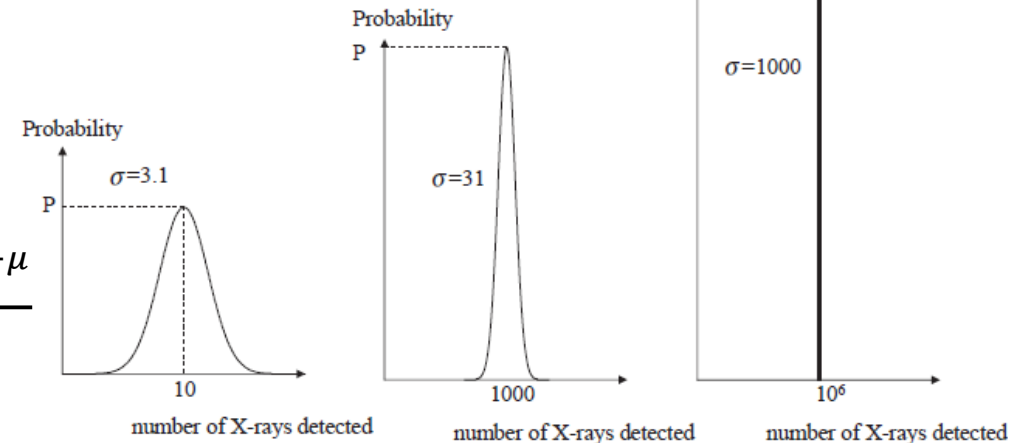


image

- Noise standard deviation, σ , in the detector is proportional to $\sqrt{N}$, N being the incident # of X-rays
- The noise arises from the statistical variation in the # of photons hitting a given region of the detector
- It follows Poisson distribution
- The source of statistical variation is (a) the rate of X-ray photon release from the anode has certain variance and (b) Compton scattering

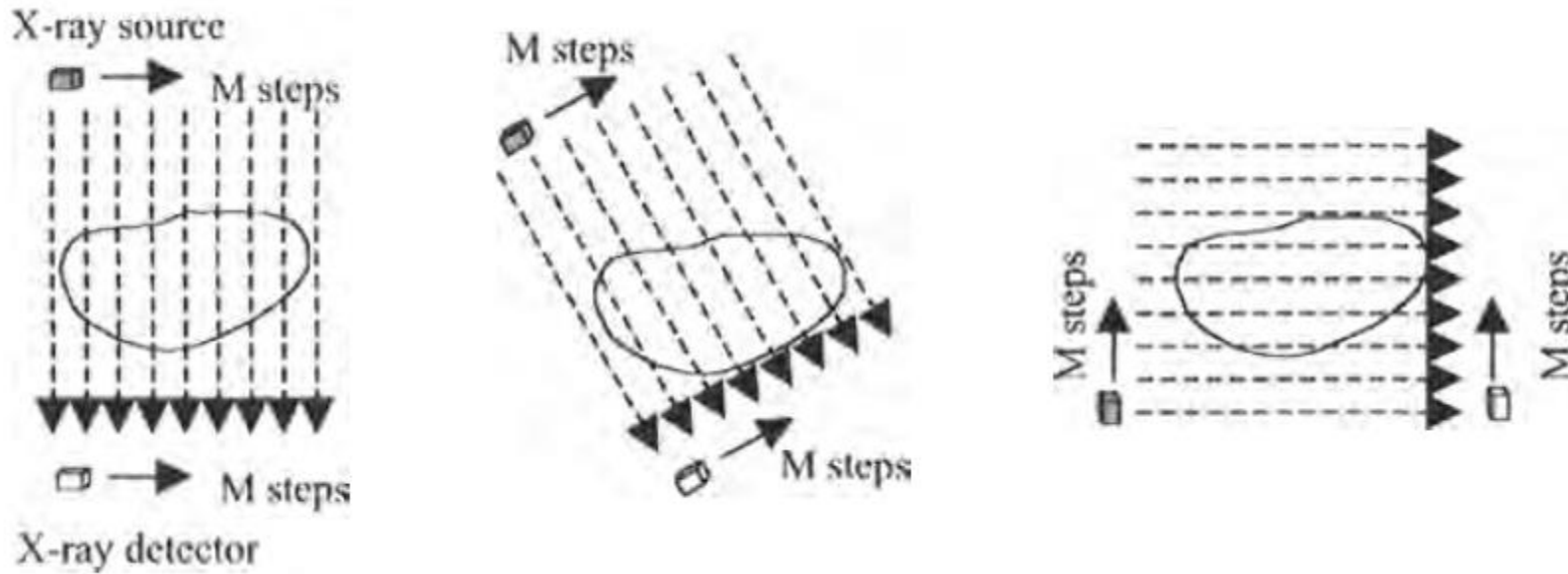
N – number of photons hitting a region
 μ – statistical mean number of photons hitting a region
 $P(N)$ probability of N photons hitting a region of the detector

$$P(N) = \frac{\mu^N e^{-\mu}}{N!}$$



Computed Tomography

First gen. CT scanners



$$I = I_0 e^{-\int \mu(x,y) ds}$$

X-ray beam is collimated in 2 dimensions for the desired slice thickness

- Tube and the detector pair moved synchronously, linearly in M distinct steps for obtaining one projection. (1D project with M points)
- Then the entire pair is rotated relative to the object and another projection was acquired ($N \cdot \theta$ projections, $\theta = 180/N$)
- $M \times N$ matrix is obtained, from which, an image is obtained using appropriate reconstruction algorithm

CT scanners – Fan Beam



third generation



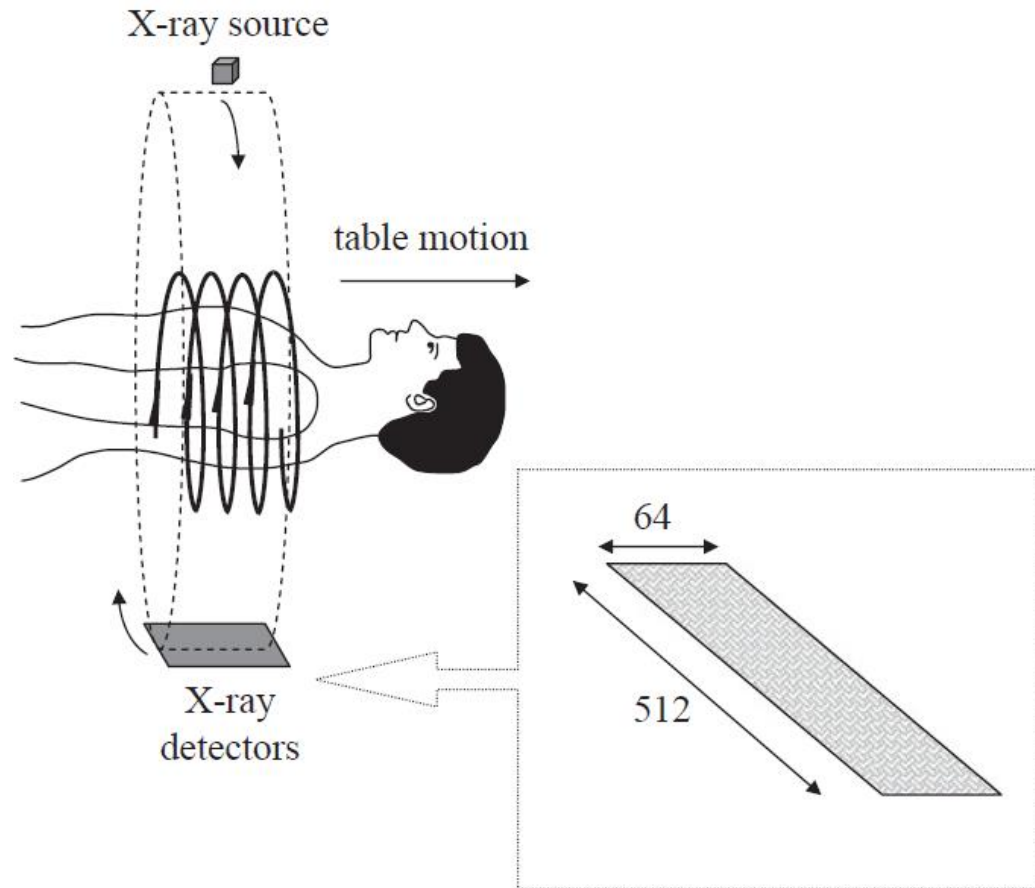
fourth generation

Third generation (currently the industry standard)

- X-ray beam shaped as a fan, through collimation in 2 dimensions for the desired slice thickness and angular spread (45° to 60°)
- Single Tube + detector array pair is rigidly mounted on an arm that rotates for acquiring multiple projects.
- Each projection is about 0.125 to 0.5 msec exposure
- One full rotation (acquiring all projects for single slice) is $\sim .25$ to 1.5 secs.

- Tube currents can be up to 1 Ampere

Multi-detector CT: Spiral or helical scanning



- Helical scanning enables the collection of multiple slice data very fast.
- The source/detector array pair rotate at the fixed gantry, as the patient table moves. This results in the X-ray source/detector pair traversing a helical path around the subject.

$$p = \frac{d}{S}$$

S – collimated slice thickness
d – table feed rate of *d* mm/rotation
p – pitch of the helical path

Typical pitch values are 0.5 to 2

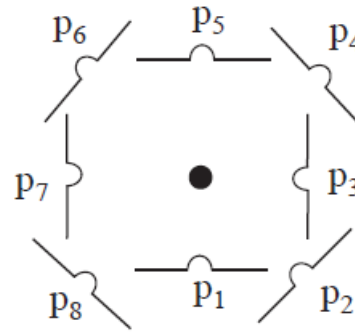
Multiple projections

Multiple projections at different angles, over 180 degrees

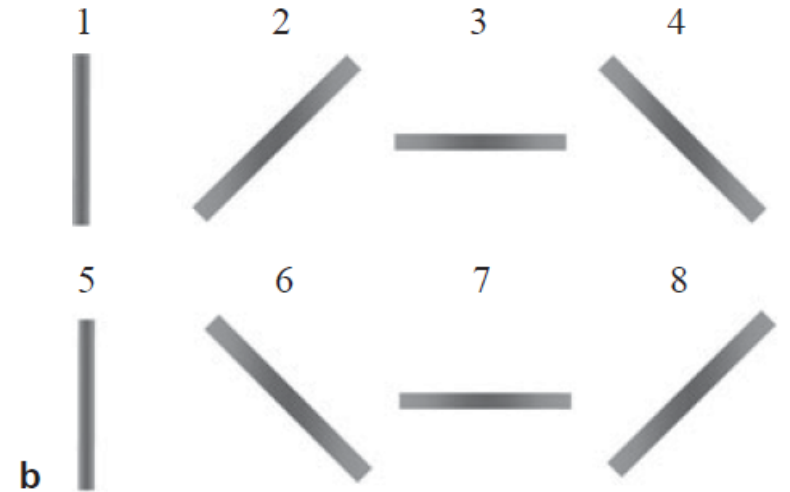
$$I = I_0 e^{\int \mu(x,y) ds}$$

Reconstruction involves mathematical operations on the acquired data

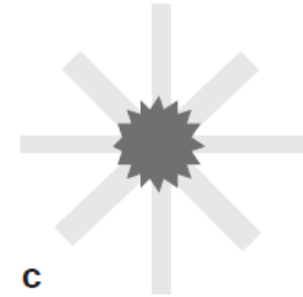
$$\mu(x,y) = \int_0^{\pi} p(x \cdot \cos \theta + y \cdot \sin \theta, \theta) \cdot d\theta$$



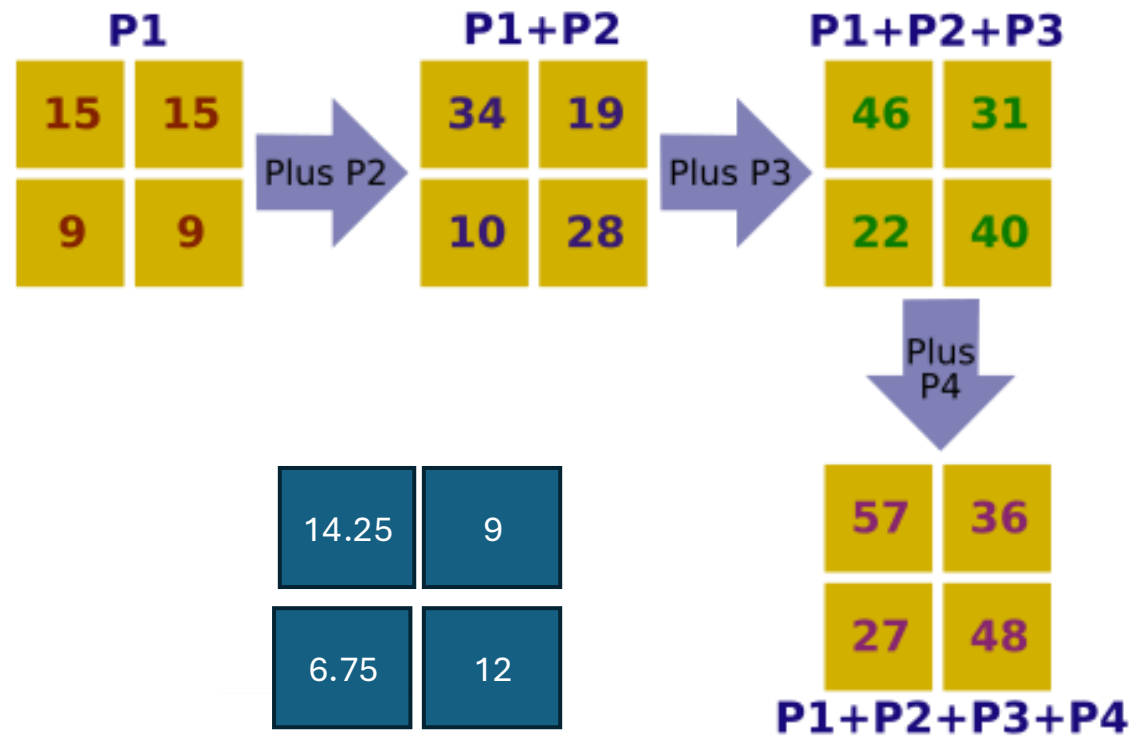
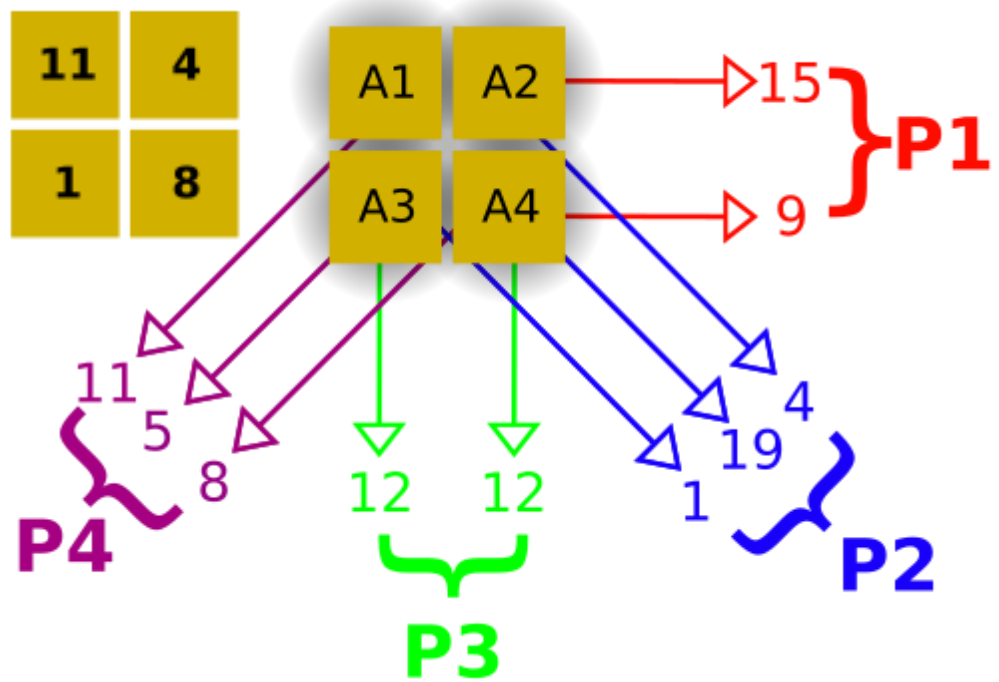
a



b



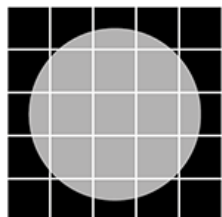
c



Division by #of
projections

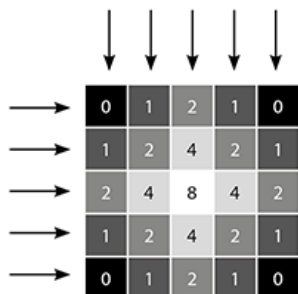
14.25	9
6.75	12

1. If we use an example of a high attenuation circular object on a matrix with each square representing a pixel



2. The attenuation value is averaged over each pixel

3. Beams are projected in two directions



4. As mentioned previously, as each beam passes through the object a summed linear attenuation coefficient (LAC) is obtained

4	0.8	0.8	0.8	0.8	0.8
10	2	2	2	2	2
20	4	4	4	4	4
10	2	2	2	2	2
4	0.8	0.8	0.8	0.8	0.8

5. We then assume that the assumed LAC is spread evenly across each beam path

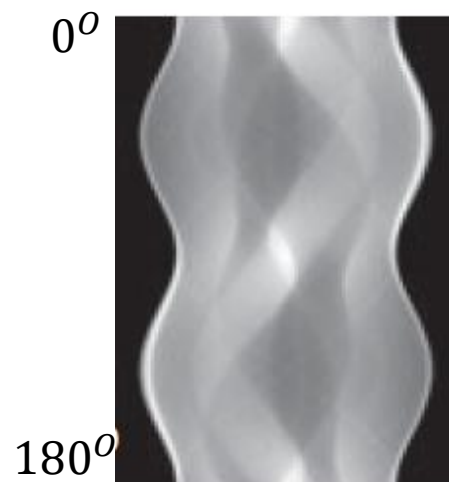
4	10	20	10	4
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0.8	2	4	2	0.8
0.8	2	4	2	0.8
0.8	2	4	2	0.8
0.8	2	4	2	0.8
0.8	2	4	2	0.8

6. Each table of calculated LACs is added. The resulting table provides an estimate of each pixel's original LAC

1.6	2.8	4.8	2.8	1.6
2.8	4	6	4	2.8
4.8	6	8	6	4.8
2.8	4	6	4	2.8
1.6	2.8	4.8	2.8	1.6

SINOGRAM



Ground truth



Output of simple back projection

Filtered Back Projection

$p(r, \theta)$

SINOGRAM

0°



180°



Ground
truth



Output of
simple back
projection

Each projection/view is filtered to remove the $1/r$ weighting.

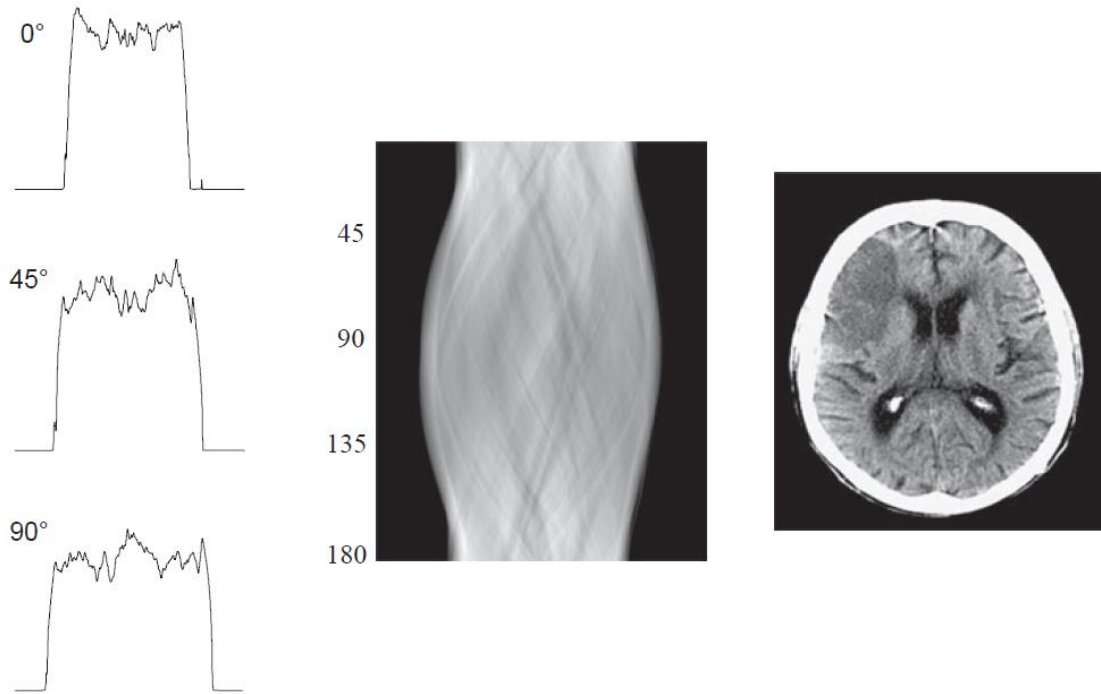
Filtering is usually done in Fourier Domain

$P(k, \theta) = 1D \text{ FT}[p(r, \theta)]$ along the r dimension

$$\hat{p}(r, \theta) = 1D \text{ FT}^{-1}[P(k, \theta) \cdot \text{Filter}]$$



CT Hounsfield Unit



What the CT image displays is not strictly $\mu(x, y)$, but is the following

$$CT_0 = 1000 \cdot \frac{\mu_0(x, y) - \mu_{H_2O}}{\mu_{H_2O}}$$

$\mu_0(x, y)$ is the tissue's linear attenuation coefficient in each pixel

CT_0 is the CT number expressed in Hounsfield Units

CT Hounsfield Unit

Table 2.1: CT numbers of different tissues at 70 keV

Tissue	CT number (Hounsfield units)
Bone	1000–3000
Muscle	10–40
Water	0
Lipid	–50 to –100
Air	–1000
Brain (white matter)	20 to 30
Brain (grey matter)	35 to 45
Blood	40

What the CT image displays is not strictly $\mu(x, y)$, but is the following

$$CT_0 = 1000 \cdot \frac{\mu_0(x, y) - \mu_{H_2O}}{\mu_{H_2O}}$$

$\mu_0(x, y)$ is the tissue's linear attenuation coefficient in each pixel

CT_0 is the CT number expressed in Hounsfield Units

Fan-Beam to Parallel projection data

- Sorting $p(r, \theta)$ data to create synthetic projections from parallel beams

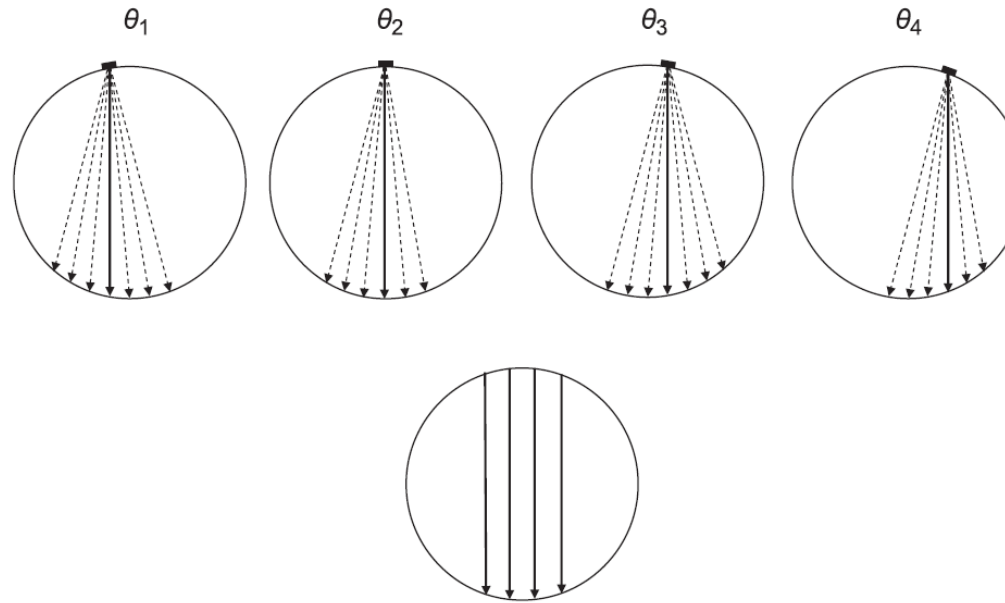


Figure 2.33

The process of data rebinning to produce parallel projections from fan beam data. Data from four different projection angles, $\theta_1 \dots \theta_4$ are shown. One projection, shown as the solid line, is taken from each of the datasets to produce a synthetic data set (bottom) with four parallel projections, which can then use a simple filtered backprojection algorithm for image reconstruction.

Interpolation in Spiral CT

- Data not acquired in between the helical passes is estimated through linear interpolation
- Interpolation weights can be adjusted to reconstruct thinner or thicker slices (compared to the collimation thickness chosen) and hence, slice thickness could be chosen retrospectively in I

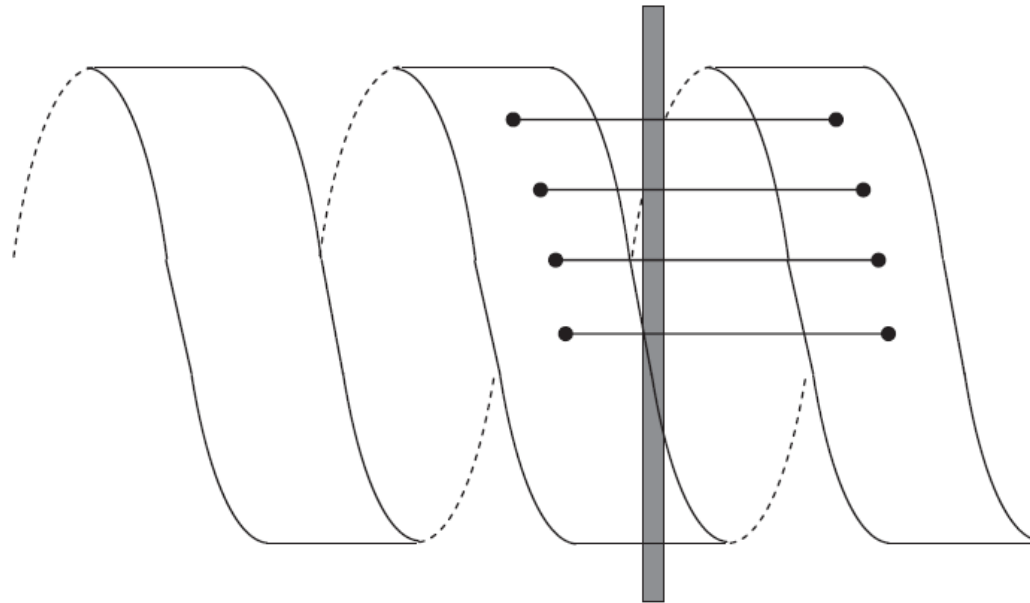


Figure 2.34

Image reconstruction from helical CT data. The process involves interpolation of data points which are not acquired. The solid dots represent acquired data, with the filled shape being the location of the interpolated data.

Beam Hardening

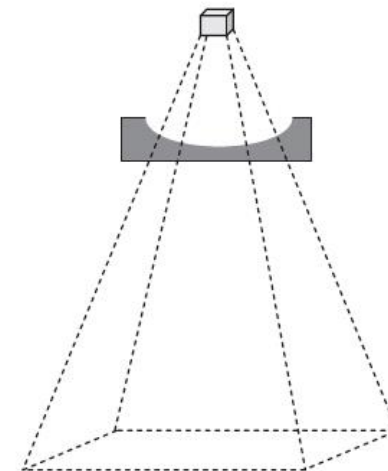
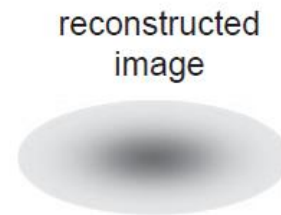
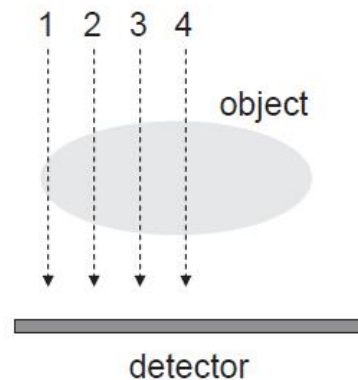
$$I = I_0 e^{\int \mu(x,E) dx}$$

$$\mu(x, E) \cdot x = \ln \left(\frac{I_0}{I} \right)$$

- The projection data $p(r, \theta)$ is assumed to be from a single energy X-ray (Monochromatic)
- However, that is not the case in reality, as the released X-rays have a spread of low and high energy rays

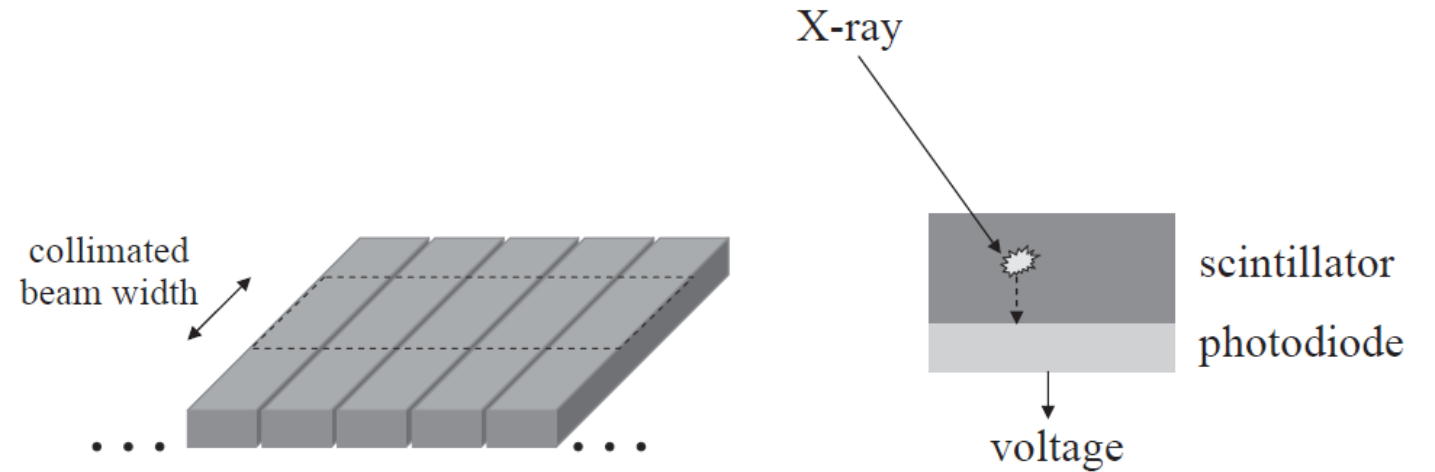
$$I = \int_0^{E_{max}} I_0(E) e^{\int \mu(x,E) dx} dE$$

$$\ln \frac{\int I_0(E) dE}{\int_0^{E_{max}} I_0(E) e^{\int \mu(x,E) dx} dE}$$



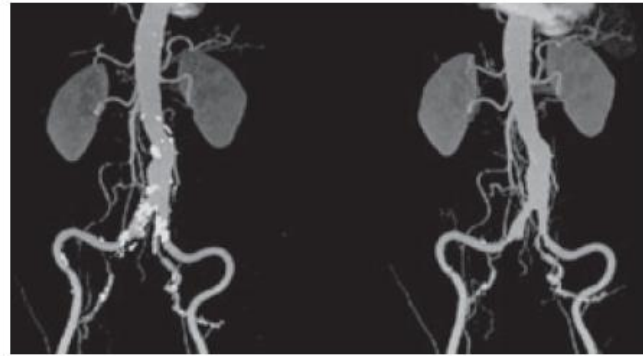
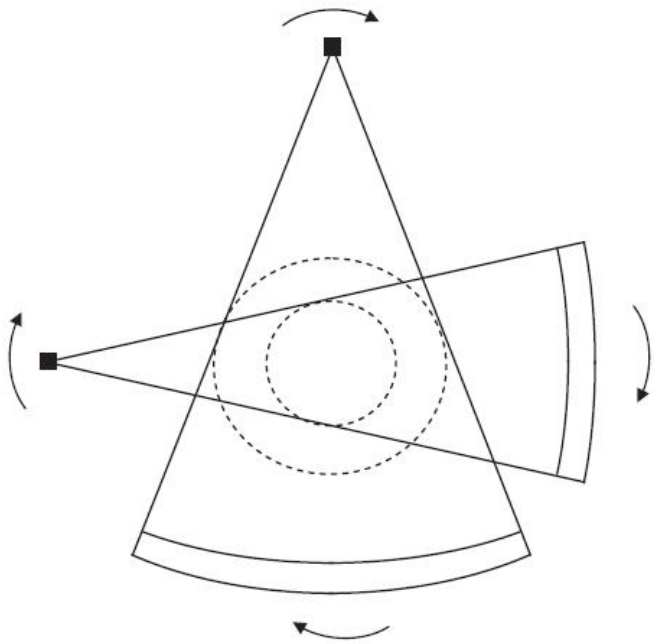
Detectors in CT

- Solid state detectors (highly sensitive, offer fast readout times and are temporally stable for multiple sequential readouts without much loss in sensitivity)
- Scintillator + photodiode



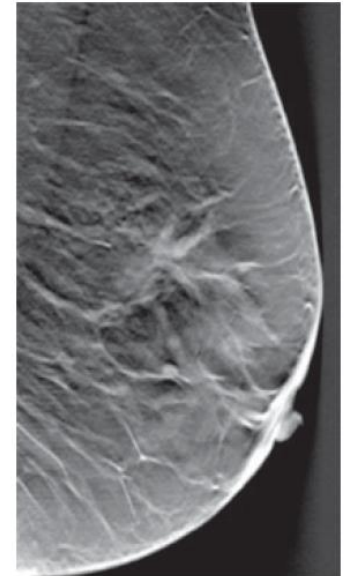
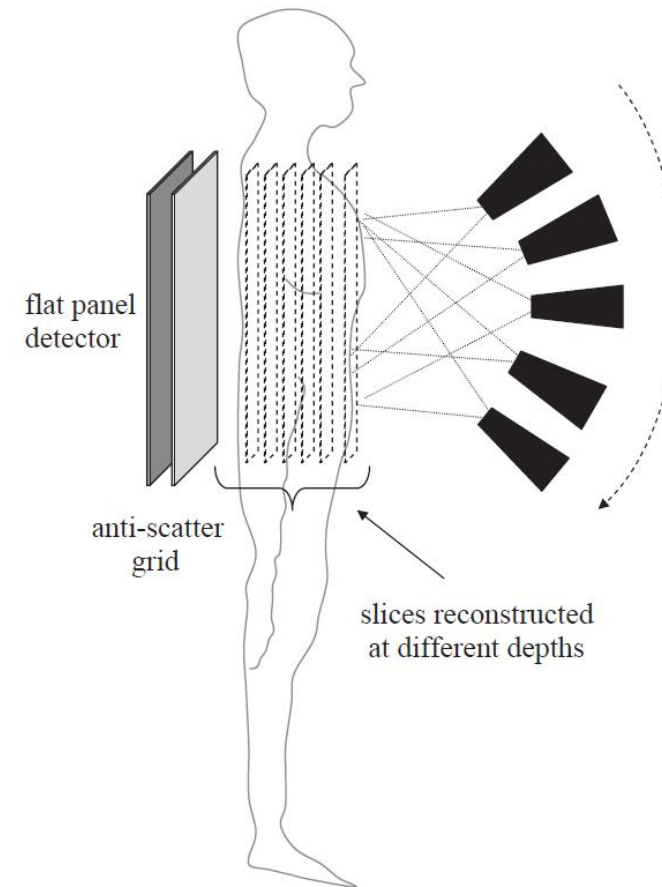
Dual Source/Energy CT

- To increase data acquisition rate (decrease scan time) or to use dual energy settings on the sources for obtaining different contrasts



Digital X-ray Tomosynthesis, Cone Beam CT

- To increase data acquisition rate (decrease scan time) or to use dual energy settings on the sources for obtaining different contrasts



Radiation Dose

Biological effects of radiation exposure are divided into two categories

- Deterministic
 - Cellular damage/death leading to loss in tissue function
 - Occurs at high radiation exposure
- Stochastic
 - Cell is alive but with the possibility of genetic mutations due to chromosomal damage (could lead to cancer)
 - Can occur at much lower dose levels, compared to deterministic effects
- Exposure/Dose (D) is measured in terms of energy (Joules) absorbed per unit weight (Kg) of the tissue.
- Units Gray or Sievert

$$D_{T,R} = \frac{1}{m_T} \int D_R dm \quad \text{J/kg – Gray(Gy) Or Sievert (Sv)}$$

Dose in X-ray imaging

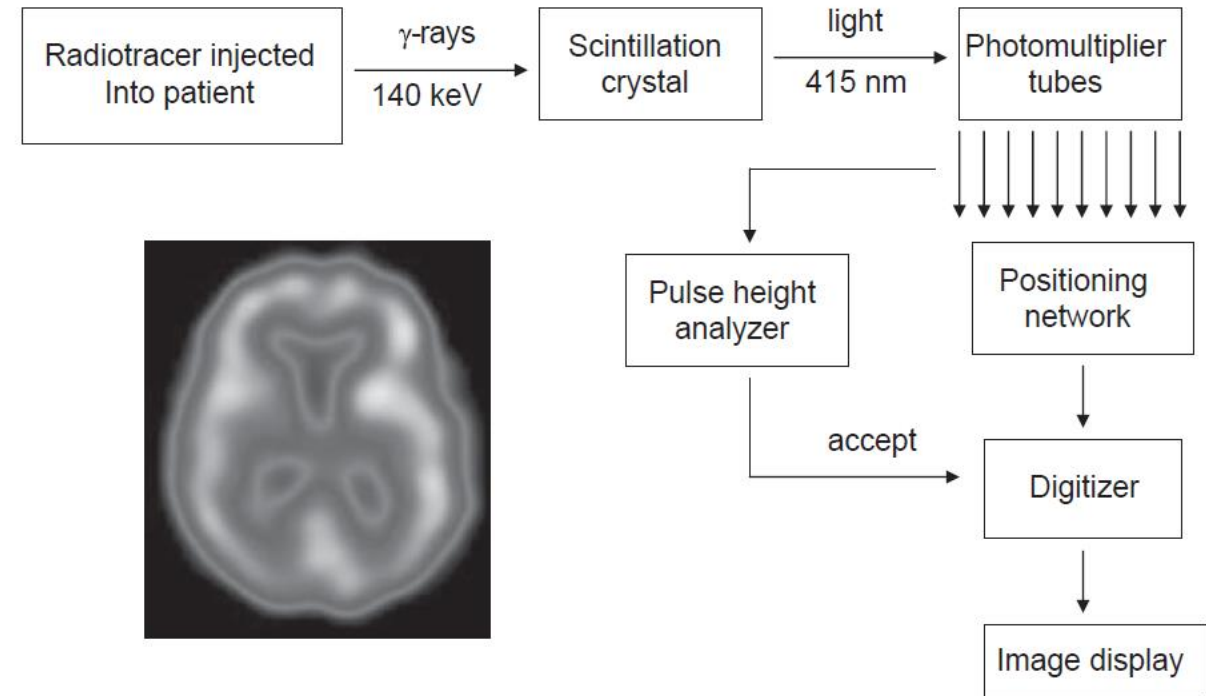
$$D_R \propto \frac{kVp^2 \cdot mA \cdot sec}{d^2}$$

$$D_R \propto \frac{(Tube\ voltage)^2 \cdot (Tube\ current - time\ product)}{(distance\ of\ the\ subject\ surface\ from\ X - ray\ Source)^2}$$

SPECT & PET

SPECT

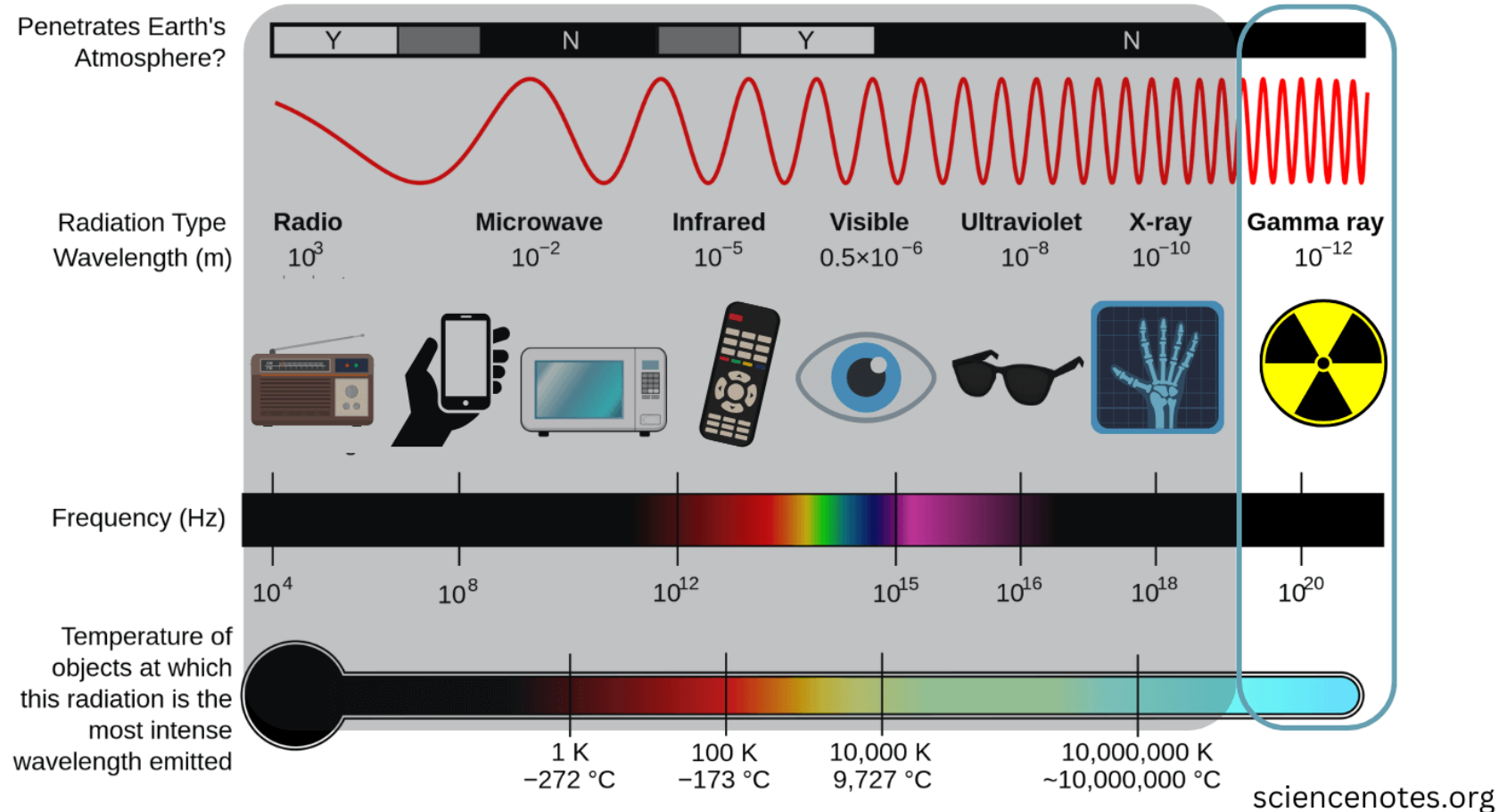
- Single Photon Emission Computed Tomography
- ^{99}Tm Chemical complexes (commonly used) – Radioactive contrast agent (Radiotracer), injected in nanograms dose.
- Release γ – rays, at single energy of 140 keV.
- 2 or more gamma cameras around a rotating circular gantry are used to capture the released gamma-ray (similar to CT based image reconstruction)
- Collimation is done so, gamma rays hitting perpendicular to the scintillation crystal are picked, avoiding scatter radiation.
- Typical Spatial resolutions – 5mm to 10mm voxel size.



SPECT

Gamma Rays or Gamma Radiation

A gamma ray is the type of electromagnetic radiation with the highest energy and shortest wavelength.



SPECT

- Half life of SPECT radiotracers – time taken for the radioactivity to reduce to half of its starting value

$$N(t) = N_0 e^{-\lambda t}$$

N is the number of radioactive nuclei at a given time.

N_0 is the # of radioactive nuclei at the beginning

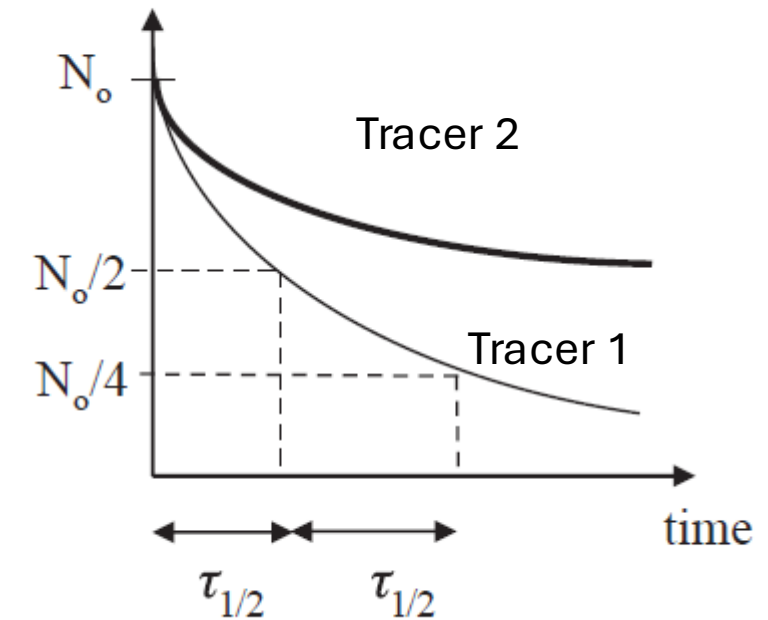
$$Q = -\frac{dN}{dt} = \lambda N$$

λ is the decay constant determining the # of nuclear disintegrations per second.

$$\tau_{1/2} = \frac{\ln 2}{\lambda}$$

Q is the radioactivity measured in Curie (Ci)

Amount of radiotracer



- Usual doses given in SPECT is in milli Curie

SPECT

- $\tau_{1/2}$ of Tm99 is 6.02 hrs.
- Radiation exposure of the patient receiving the radiotracer depends also on the rate of the excretion of the radiotracer from the body.
- Excretion rate is also exponential decay and hence can be characterised by its own $\tau_{1/2,bio}$.
- The effective half life of the radioactivity in the body is a combination of $\tau_{1/2}$ and $\tau_{1/2,bio}$.

$$\tau_{1/2,eff} = \frac{\tau_{1/2} \cdot \tau_{1/2,bio}}{\tau_{1/2} + \tau_{1/2,bio}}$$

Table 3.1: Properties of common radiotracers used in planar scintigraphy and SPECT

Radiotracer	Half-life (hours)	γ -ray energy (keV)	Clinical application
^{99m}Tc	6.0	140	various
^{67}Ga	76.8	93, 185, 300, 394	tumour detection
^{201}Tl	72	167, 68–82 (X-rays)	myocardial viability
^{133}Xe	127.2	81	lung ventilation
^{111}In	67.2	171, 245	inflammation

SPECT

- Radioactive Tm99 is often produced on-site using a Technitium generator module which is periodically (weekly) replenished.
- The Tm99 is liganded with a chemical complex that protects it from fragmentation within the body and provides selectivity for specific target organ (bone repair, neuroreceptors or blood flow)



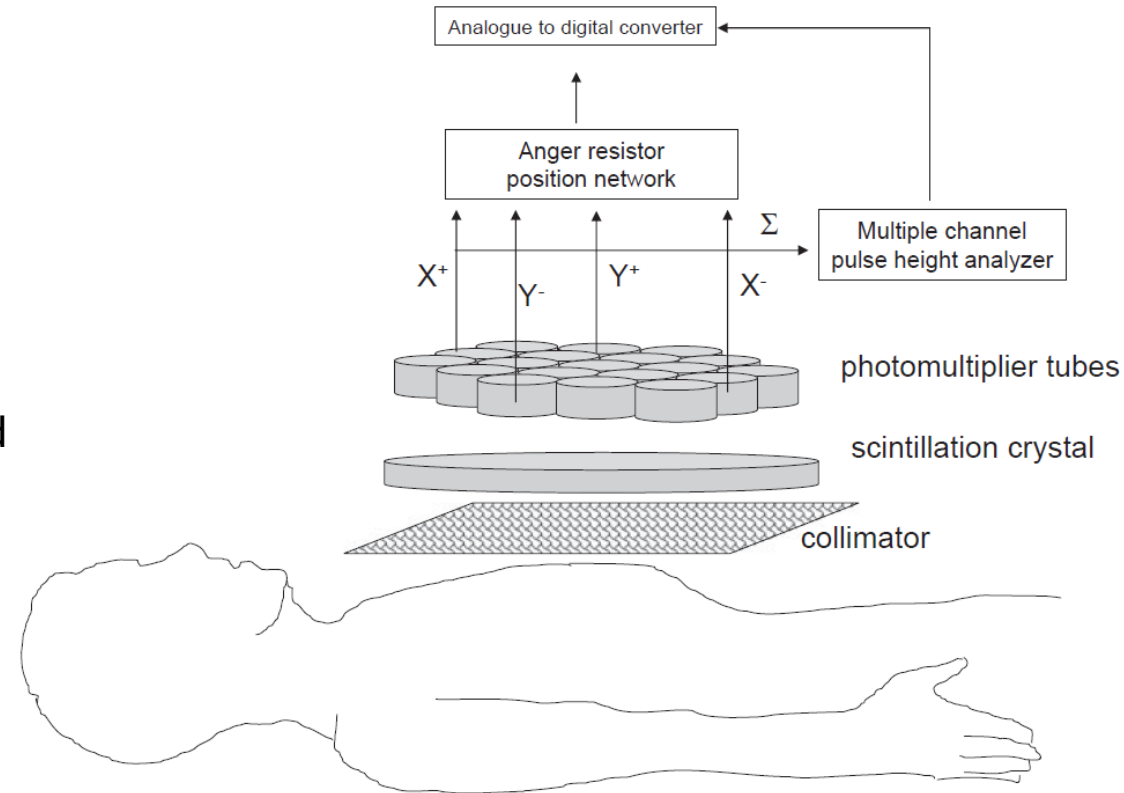
Components of a SPECT System

A SPECT system consists of:

- Collimator
- Scintillation crystal (NaI(Tl))
- Photomultiplier tubes (PMTs)
- Positioning electronics
- Energy window analyzer
- Rotating gantry
- Reconstruction computer

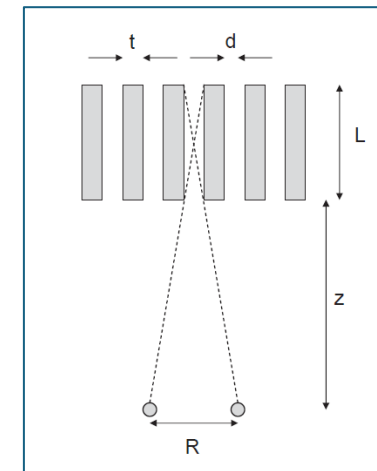
The Gamma Camera

- Needs to be able to detect $>10000 \gamma$ – rays per second.
- Is positioned close to the patient for improved SNR
- NaI(Tl) scintillation crystals. Thallium doped Sodium Iodide
- 99% of γ radiation gets absorbed in the body or scattered and do not reach the detector.
- Should have sufficient energy resolution to reject scattered γ rays.
- Collimation plays an important role in determining resolution.



$$\text{Resolution OR voxelsize, } R \approx \frac{d(L + z)}{L} \quad g \approx K^2 \frac{d^2}{L^2} \cdot \frac{d^2}{(d + t)^2}$$

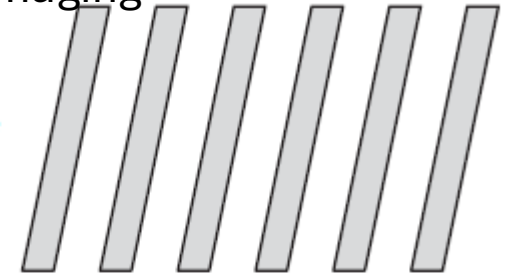
- Collimator's efficiency (g), which measures the fraction of gamma rays detected out of the total emitted, is another parameter that affects Image quality (affects SNR)



The Gamma Camera

Cardiac or breast imaging

slanthole



Magnifying effect

converging



Large FOV

diverging



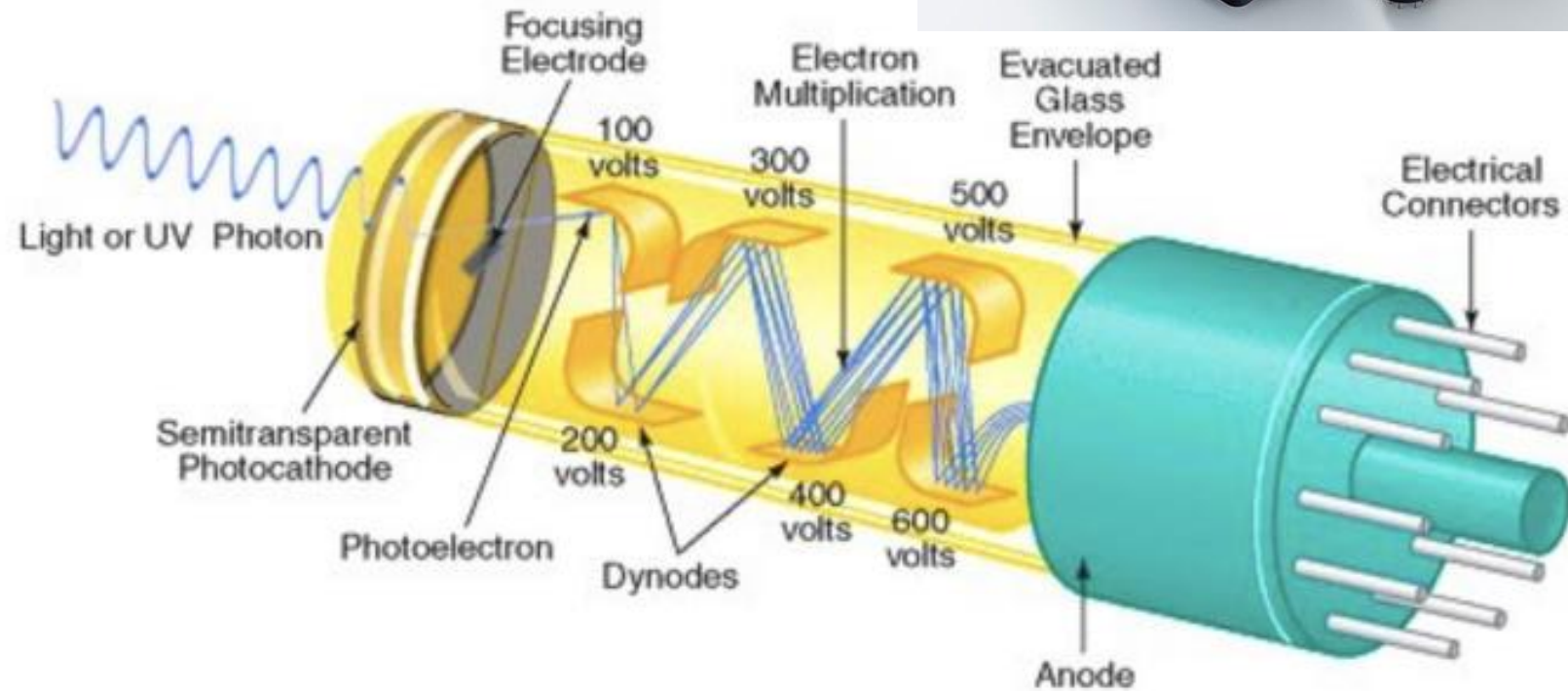
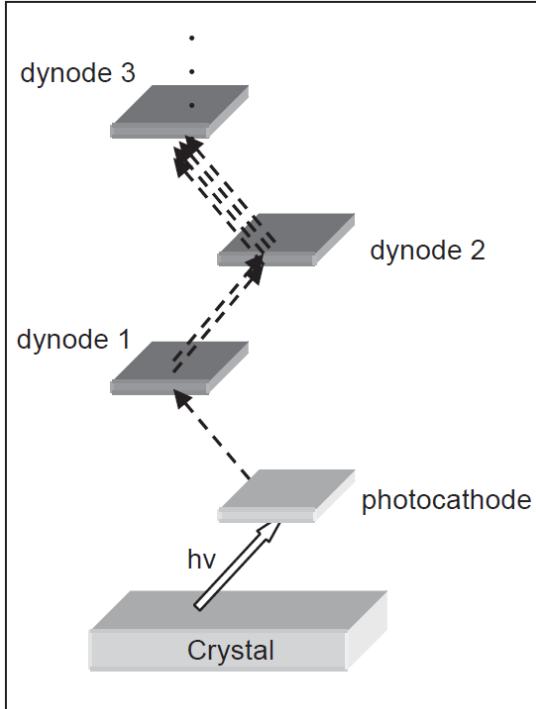
Magnifying for Very small
organs like thyroid

converging

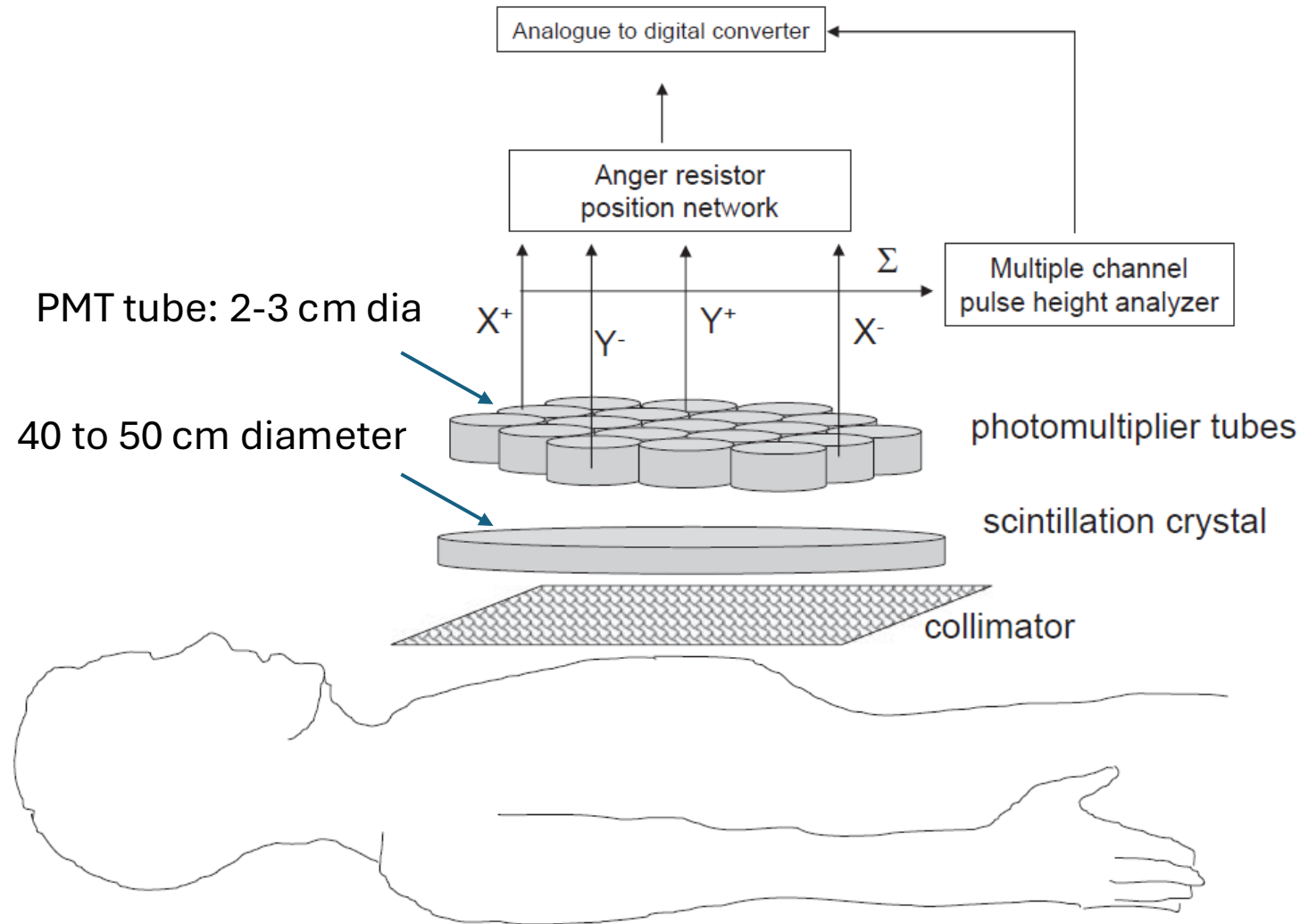


The photo multiplier tube

- CsSb – Cesium Antimony coating in photocathode and dynodes
- Amplification factor of 10^5 to 10^7



Scintillator + PMT



- Single large scintillator disc is usually used.
- Discretized scintillators are also coming up in newer designs.
- Need to be completely sealed, for NaI(Tl) crystals as they are hygroscopic (absorbs water)
- Differences in the responses of PMT tubes for a given gamma ray (non idealities) need to be corrected for through calibration.
- Positioning network that helps in further localizing the location of the scintillation event in X-Y plane – improves the resolution to 3 - 5 mm.

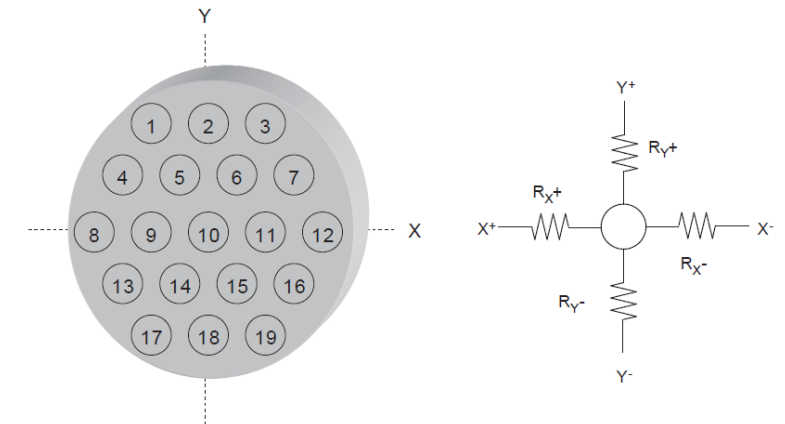
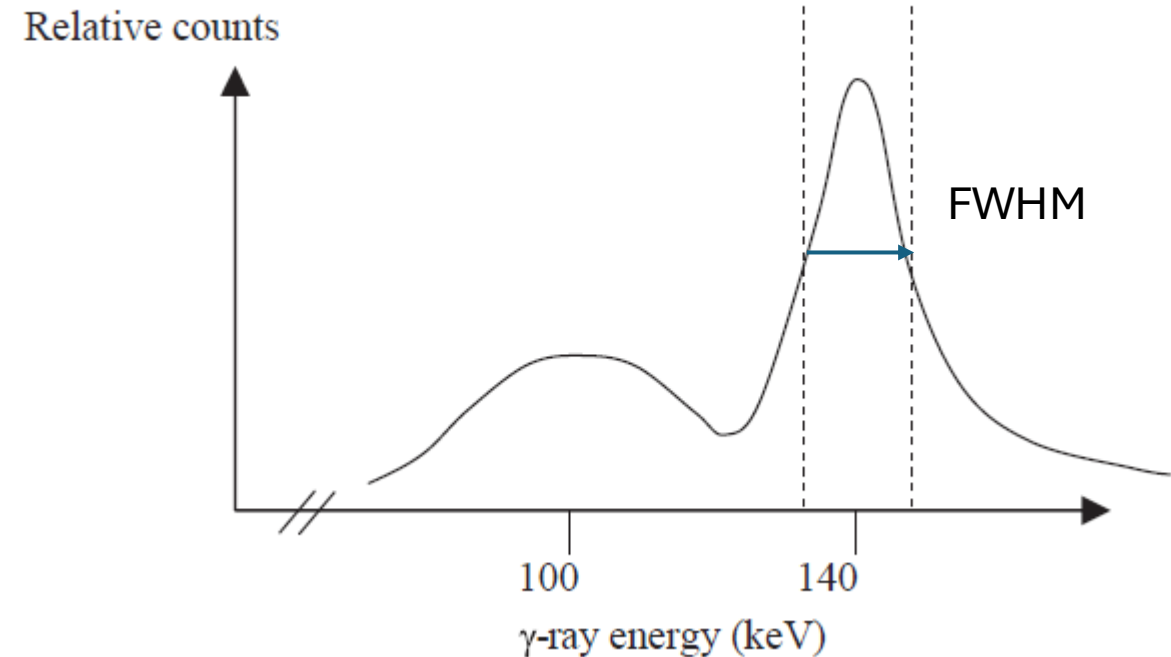


Figure 3.9

(left) Nineteen PMTs in a hexagonal arrangement on an NaI(Tl) crystal. (right) The four-resistor network attached to each PMT.

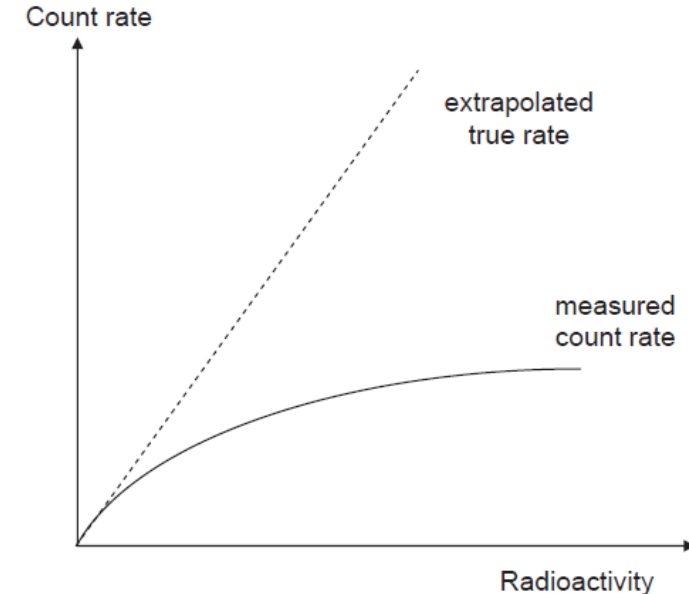
Energy resolution of the Scintillator

- Incident γ rays pertain to a spectrum of energies due to scattering within the body.
- The narrower the FWHM of the scintillator for the specific energy, the better is the discriminatory capacity of the scintillator.
- Pulse Height Analyser (PMT response is proportional to the energy of the incident ray) is used to discriminate scintillation events of the desired energy range from lower energy γ rays.



Instrument dead time (Response time of the system)

- The time during which the system is not responding to a new gamma ray hitting the scintillator; Paralysable (variable time) or non-paralysable (fixed time)
- Reasons:
 - Paralysable: It takes about 230nsecs after a gamma-ray hits the scintillator, to release photons corresponding to that event.
 - If another gamma ray hits the scintillator before within this 230 nsecs, the next release of light photons gets delayed by another 230nsecs, but the net amount of released photons is double.
 - Hence even though two events occurred, it is counted as 1. Extend this to multiple gamma rays hitting one after another within the 230 nsecs...the dead time keeps extending as more number of gamma rays hit back to back...thus underestimating the # of counts or recorded events (consequently concentration of the radiotracer).
 - Non paralysable: Once an event occurs, it takes the electronics a fixed amount of time to record and reset. (PHA, positioning network etc).



Typical count rates range from 10000 to 40000 counts per sec; can go upto 700000

Image characteristics

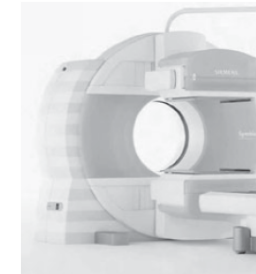
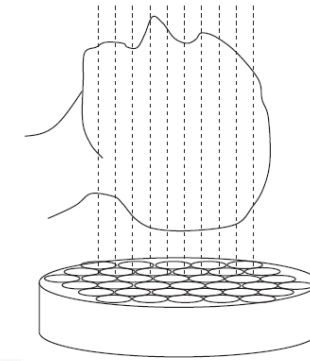
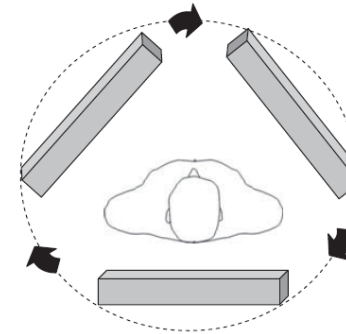
SNR

- Amount of radio tracer injected
- Time spent at each projection collecting the counts (typically a min of 100000 counts per image)
- Post acquisition filtering

Resolution

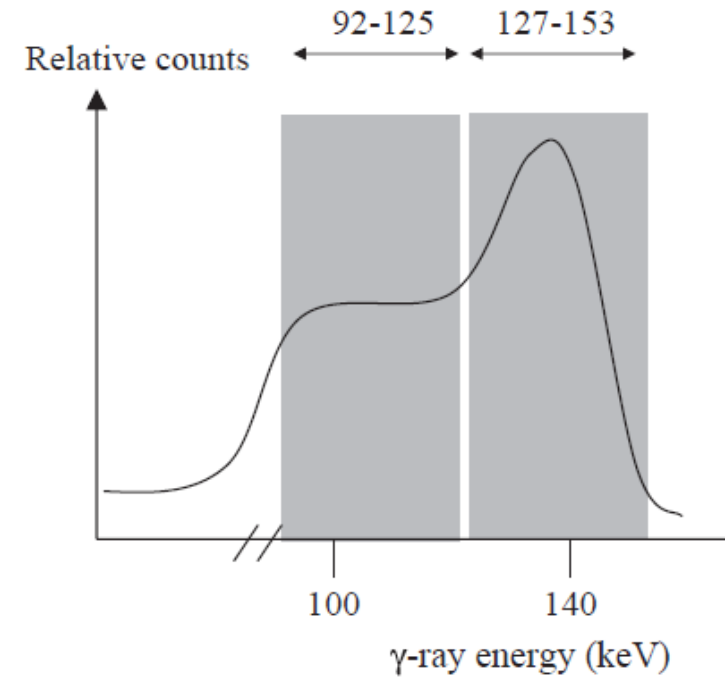
- $R = \sqrt{R_{\text{gamma}}^2 + R_{\text{collimator}}^2}$
- R_{gamma} is determined by the density of the PMTs and accuracy of positioning network
- $R_{\text{collimator}}$ – collimator characteristics
- Typical resolution is 1 to 2cms for deep organs and 5 to 8 mm for superficial regions.

Usually 360 degree projections

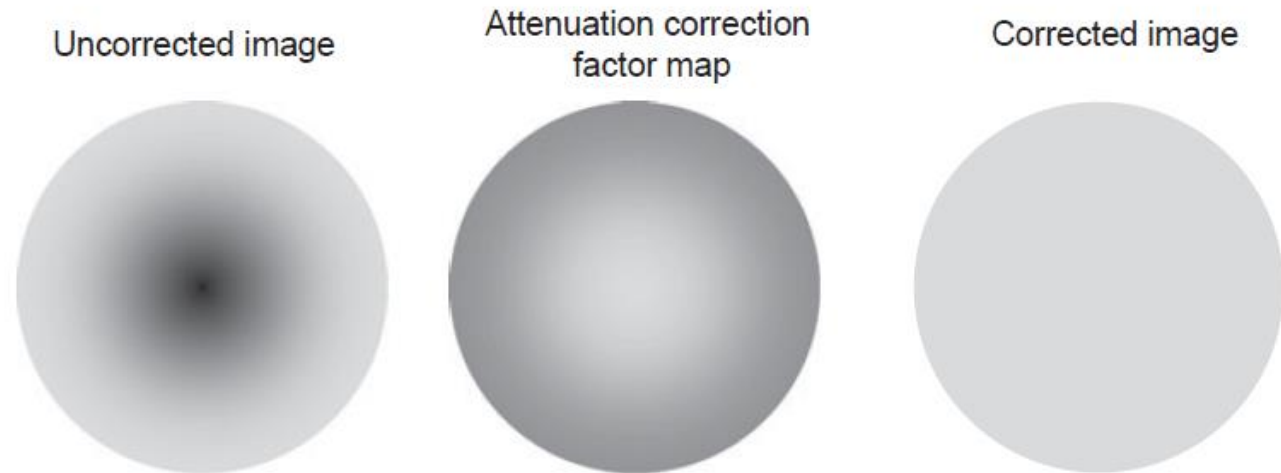


Corrections

- Scatter corrections



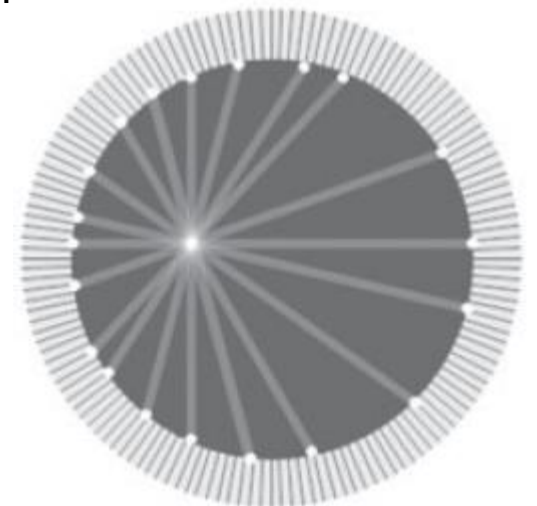
- Attenuation Corrections



PET – Positron Emission Tomography

PET

- PET Radiotracers emit positrons.
- The emitted positrons get annihilated when they hit an electron in the tissue, releasing two γ -rays in opposite directions along a straight line (line of reconstruction). The positrons travel about 0.1mm to 2mm in the tissue before getting annihilated
- The emitted gamma rays are at 511 keV, much higher than the typical tracers used in SPECT.
- Because of the intrinsic line-of reconstruction, no collimation is used in PET in reconstruction
- PET has 100 to 1000 times more SNR than SPECT because:
 - No collimation
 - Higher energy rays, so less attenuation
 - Complete ring detectors are used for PET image collection
- Require a cyclotron for generating tracers that emit positrons



PET – Radio tracers

PET tracers are usually structural analogues of biologically active molecules whose one or more atoms are replaced with a radioactive atom.

Their half lives are on the order of minutes and hence are usually generated on-site using a cyclotron.

Most pet scans are FDG tracer based.

Radionuclide	Half-life (minutes)	Radiotracer	Clinical applications
^{18}F	109.7	^{18}F FDG	oncology, inflammation, cardiac viability
^{11}C	20.4	^{11}C -palmitate	cardiac metabolism
^{15}O	2.07	H_2^{15}O	cerebral blood flow
^{13}N	9.96	$^{13}\text{NH}_3$	cardiac blood flow
^{82}Rb	1.27	$^{82}\text{RbCl}_2$	cardiac perfusion

PET scanner architecture

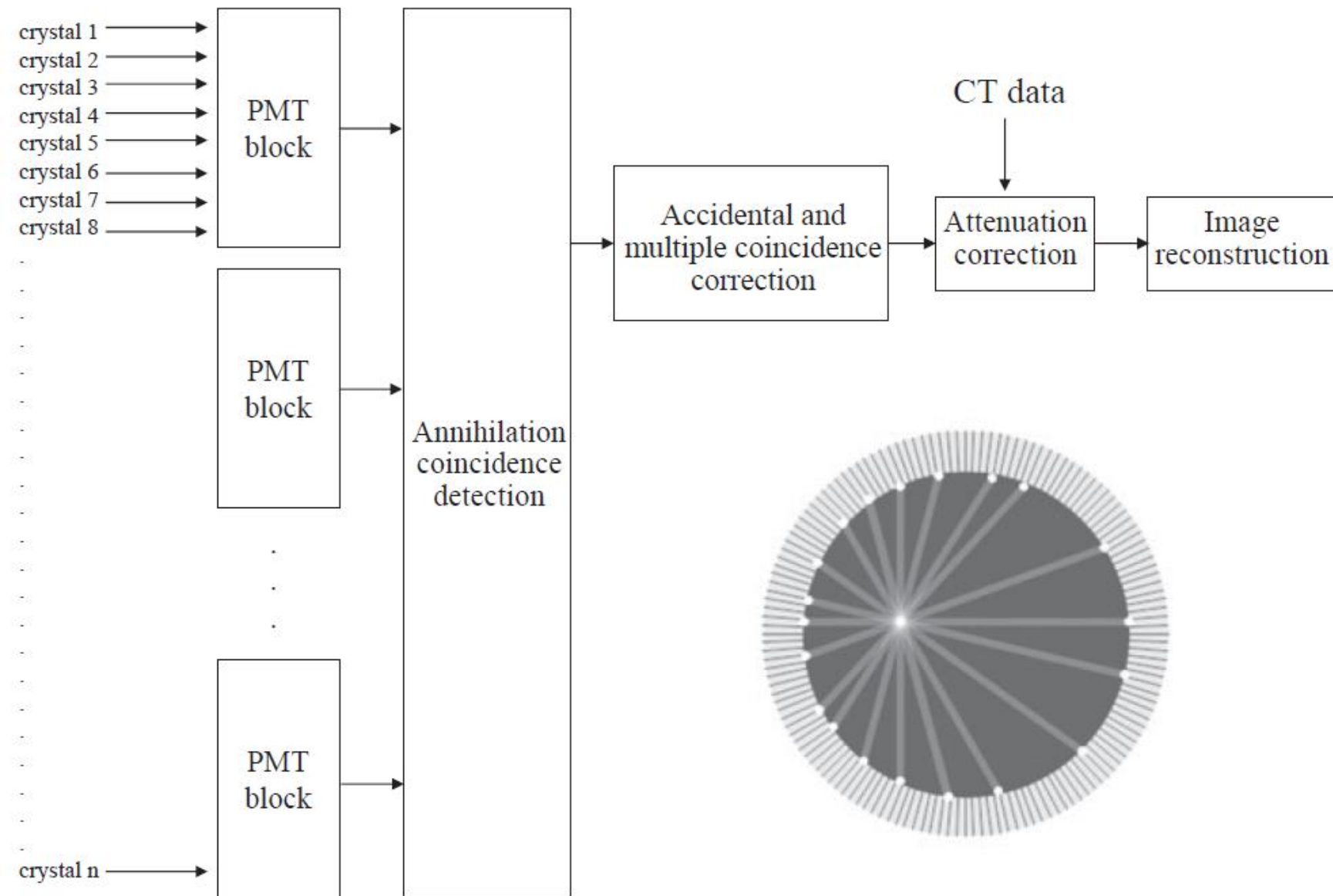
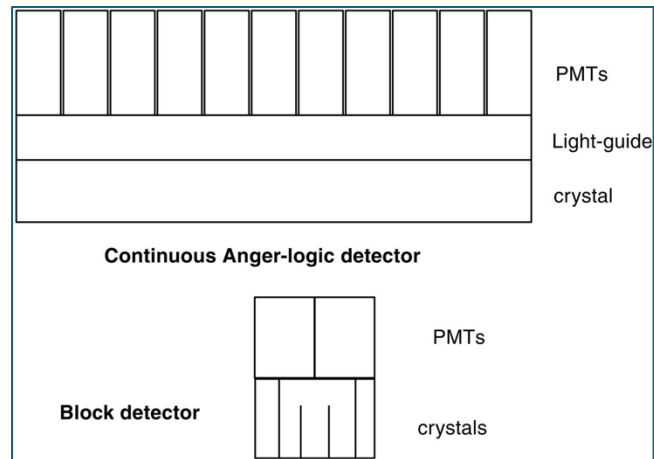
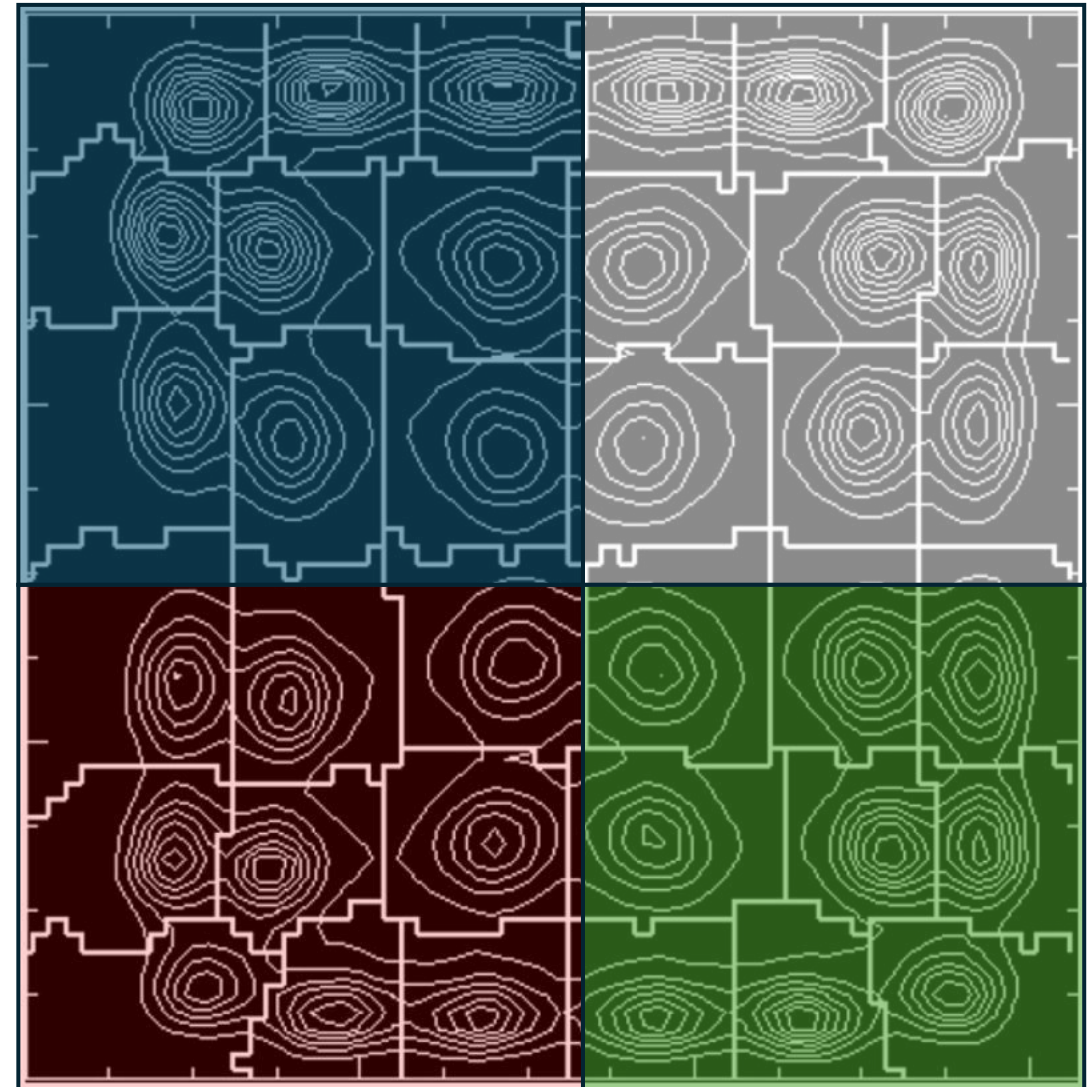


Figure 3.22

(top) The elements of a PET/CT system. (inset) Formation of lines-of-response in the PET detector ring.

PET Detector Block

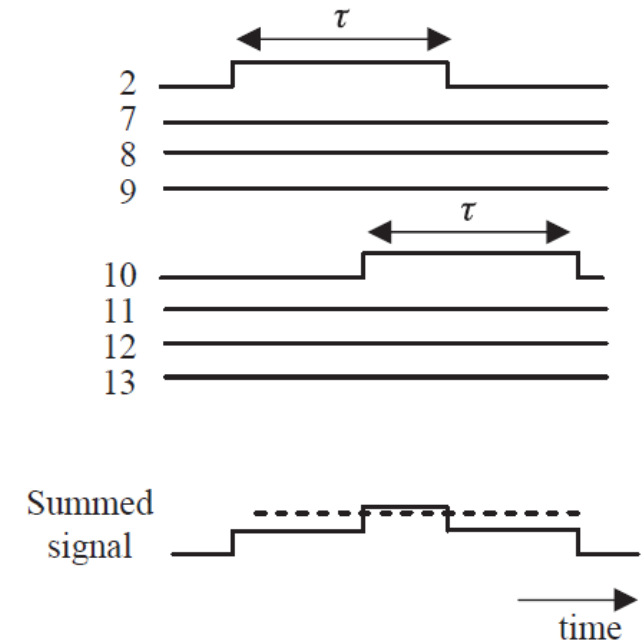
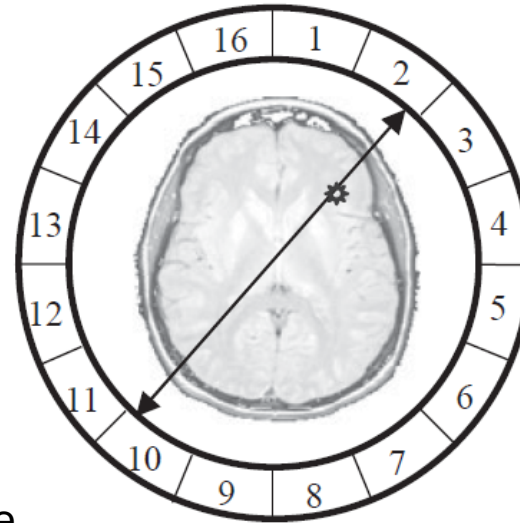
- In a block detector - typically 4 PMTs are coupled with a 2D detector array matrix of 6 x 6 or 14 x 14 matrix or 16 x 16.
- The channel cuts between the array blocks allows light to spread from one crystal to surrounding crystals. Thus an event is typically 'seen' by 4 PMTs with varying intensities.
- Anger Logic is used for position determination – i.e., determining in which crystal block did the scintillation event occur?
- Multiple such blocks are arranged in a circular ring (PET Ring) for PET detection and image reconstruction.
- Multiple such rings stacked one after another in axial direction provide large field of coverage along H-F direction



Scintillator crystal block and PMT arrangement

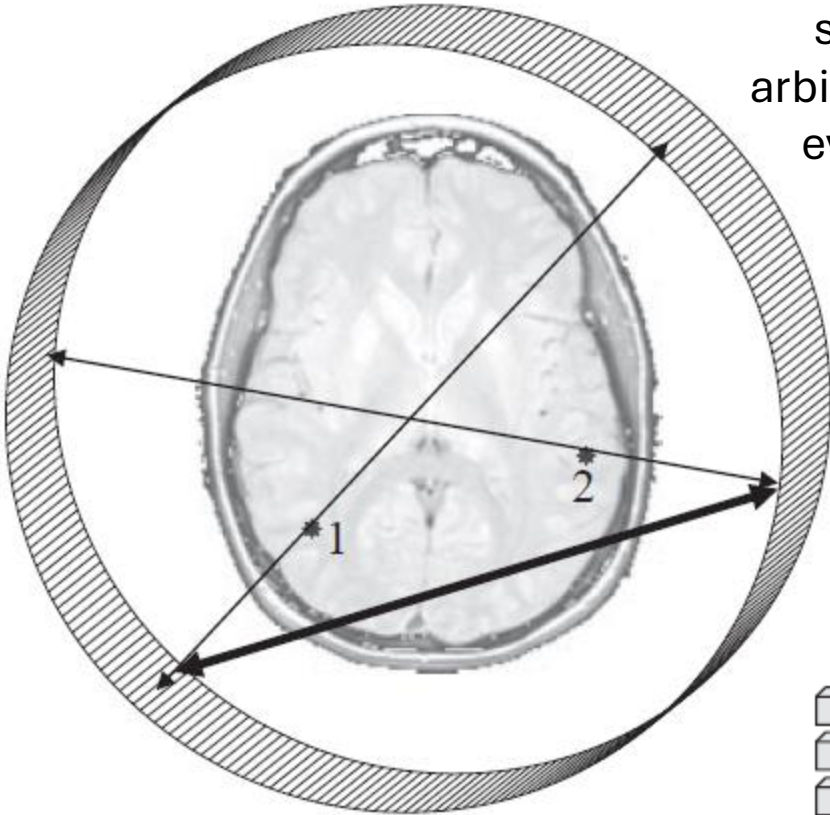
Annihilation coincidence detector (ACD)

- Each detection event leads to a voltage spike at the end of the PMT.
- If this is within the threshold window (for filtering out scatter radiation), the pulse height analyser generates a 'logic pulse', which is typically 6 to 10 nsecs long.
- Each detection event – defined based on the logic pulse generation – is given a time-stamp (within 1 to 1nsecs tolerance).
- Because of the paired release config – for each detector an opposite coincident detector exists.
- An event is considered coincident by the ACD, if the sum of the logic pulses generated from the detector pair, within the coincidence resolving time is greater than a threshold value.
- Coincidence resolving time is twice the logic pulse length,

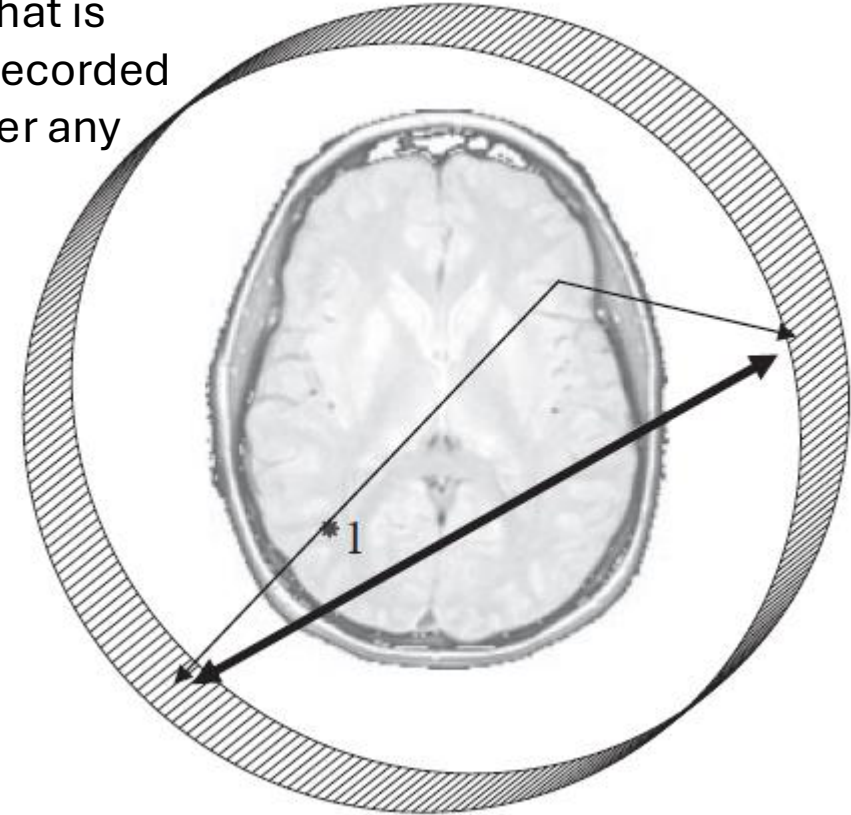
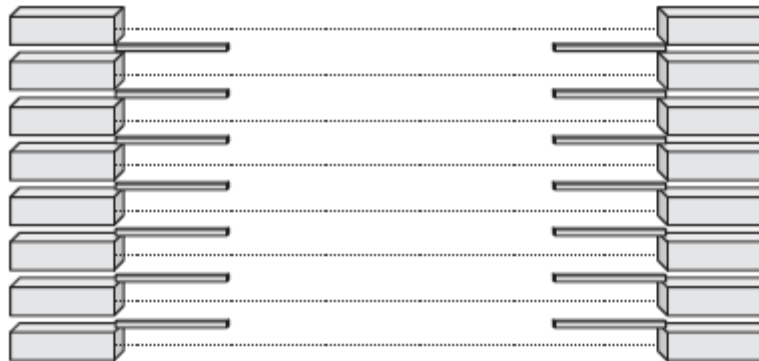


Accidental coincidence and wrong mapping

Random coincidences are recorded by using a second coincidence timing window, that is arbitrarily at a much later time than any recorded event at a detector pair. (~60nsecs after any actual event)



Collimation between rings
2D direct planes



Time of flight detection

- Time difference between the detection of simultaneous events helps in reducing the line-of-reconstruction

$$\Delta x = \frac{c\Delta t}{2},$$

